Journal Pre-proof



The statistical overloading framework for accurate evaluation of pollutant dispersal with rigorous uncertainty estimation

K.A. Krishnaprasad, R. Patel, C. El Khoury, A.J. Banko, N. Zgheib, S. Balachandar

PII: S0021-8502(25)00067-9
DOI: https://doi.org/10.1016/j.jaerosci.2025.106590
Reference: AS 106590

To appear in: *Journal of Aerosol Science*

Received date : 24 February 2025
Revised date : 11 April 2025
Accepted date : 13 April 2025

Please cite this article as: K.A. Krishnaprasad, R. Patel, C.E. Khoury et al., The statistical overloading framework for accurate evaluation of pollutant dispersal with rigorous uncertainty estimation. *Journal of Aerosol Science* (2025), doi: https://doi.org/10.1016/j.jaerosci.2025.106590.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2025 Published by Elsevier Ltd.

Journal Pre-proof

Revised Manuscript (text UNmarked)

Click here to view linked References

The statistical overloading framework for accurate evaluation of pollutant dispersal with rigorous uncertainty estimation

K.A. Krishnaprasad[a,*], R. Patel[a], C. El Khoury[b], A.J. Banko[c], N. Zgheib[d,e], S. Balachandar[a,*]

[a]*Department of Mechanical and Aerospace Engineering, University of Florida, Gainesville, 32611, FL, USA*
[b]*Department of Mechanical Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia*
[c]*Civil & Mechanical Engineering, United States Military Academy, West Point, 10996-1905, NY, USA*
[d]*Department of Mechanical Engineering, The University of Texas Rio Grande Valley, Edinburg , 78573, TX, USA*
[e]*The University of Texas Rio Grande Valley Center for Advanced Manufacturing Innovation and Cyber Systems (CAMICS), Edinburg, 78573, TX, USA*

Abstract

Accurate characterization of turbulent dispersal of aerosols and pollutants is a topic of interest involving turbulent flows in a variety of indoor and outdoor settings. For the case of a ventilated indoor space, the stochastic nature of the dispersal process results in variations due to factors such as turbulent transport and spatial inhomogeneity. Statistical overloading is a novel technique wherein the computational domain is overloaded with an abundance of pollutant particles that are randomly seeded over space and time. This would allow us to capture quantities of interest, such as mean and variation of pollutant concentration, to any desired accuracy for all possible pollutant release and sensing locations, using just one master simulation. In this study, the statistical overloading framework is employed for the case of turbulent dispersal in ventilated indoor spaces using Euler-Lagrange LES simulations in a canonical room of dimensions $10 \times 10 \times 3.2\,\mathrm{m}^3$. We establish clear guidelines for selecting computational parameters involved in designing turbulent dispersal simulations, with potential applications to other challenges involving aerosol, particulate, or pollutant dispersal. These parameters include, but are

*Corresponding authors
Email addresses: krishnap.kalivel@ufl.edu (K.A. Krishnaprasad), bala1s@ufl.edu (S. Balachandar)

Journal Pre-proof

not limited to, the minimum number of particles to be tracked during the simulation and the minimum number of turbulent/spatial realizations required to achieve converged statistics to any specified level of accuracy. We leverage the extensive Lagrangian statistics obtained from Euler-Lagrange simulations, combined with well-established statistical theory, to derive the aforementioned guidelines and requirements.

Keywords: turbulent dispersal, statistical overloading, aerosols

1. Introduction

Accurately modeling turbulent transport and pollutant dispersal is essential for evaluating exposure risks and developing effective mitigation strategies across various settings. In indoor settings, predicting the spatial and temporal patterns of turbulent dispersal is important in determining the risk of exposure to airborne diseases, chemical contaminants, or pollutants [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]. The increasing mortality rate due to inhaling fine particulates in urban environments, as reported by Schwartz and Dockery [12], is also a cause for concern. These particulates could originate from sources such as automobile exhaust, industrial smokestacks, or the accidental release of harmful substances, particularly in densely populated urban areas [13, 14, 15, 16, 17]. Assessing the risk of exposure and mitigation strategies for such scenarios are of utmost importance. In addition to the examples mentioned above, modeling the transport and deposition of fine particles has diverse applications, including drug delivery [18] and atmospheric dispersal of tephra [19].

The turbulent dispersal of a pollutant between two points can be characterized by a source, where the pollutant is released, and a sink, where the exposure level is measured. The cumulative exposure to a pollutant at the sink is significantly influenced by three factors: the duration of pollutant release at the source, the length of time the sink is exposed to the pollutant, and the time lag between these two periods. Furthermore, it is important to recognize turbulent dispersal as a chaotic process. Consider the scenario of an accidental release of pollutant from a source. The resulting pollutant concentration at the sink location is a stochastic function of time that depends on the direction of the prevailing wind from the source towards the sink. The packets of air that leave from the source containing the pollutant particles arrive at the sink in a random manner dictated by the turbulent flow. As a result, the level of exposure at the sink will vary from one run of the experiment to another [20, 21]. If we define the mean exposure as the average over many repetitions of the experiment, the variability in the exposure observed in any single run of the experiment, about the mean exposure, can be substantial; this we refer

Journal Pre-proof

to as *variability due to turbulence.*

An additional challenge in addressing pollutant dispersal is the absence of prior knowledge or the uncertainty regarding the location of the source [22, 23, 24, 25]. Similarly, the location of the infected person (or the sink) is often not known. Nevertheless, guidelines have been drawn based on average scenarios. For example, consider the 2 m social distancing guideline recommended within indoor spaces during the pandemic [26, 27]. The sick (source) and the susceptible (sink) individuals can be located 2 m apart within a room in many different ways, especially in relation to the air circulation within the room driven by the air-conditioning system [22, 28]. Clearly, the risk of exposure to the pollutant can show substantial spatial variation, which we refer to as *variability due to inhomogeneity.*

The well-mixed model [29, 30] is often used to predict pollutant concentration in indoor settings. Although it ignores the effect of inhomogeneity, its huge popularity is due to its demonstrated ability to successfully predict the room-averaged pollutant concentration accurately [22]. The importance of accounting for the dependence on source-sink separation distance has now been well recognized, and there have been studies that improve upon the well-mixed model [10, 31, 32]. The role of inhomogeneity in promoting variations about the well-mixed average has, however, not been investigated thoroughly. The problem of pollutant dispersal in a turbulent environment must therefore be approached as a two-point two-time inhomogeneous stochastic process.

It must be emphasized that the focus of the present work is on airborne spreading of pollutant. In the case of viral infection, other modes of spreading, such as through fomite [26], must also be considered. Limiting attention to airborne spreading, in order to reliably estimate the quantitative level of exposure to the pollutant at the sink, in a scenario of accidental release, the following pieces of information are needed at a minimum. (i) *The mean cumulative pollutant concentration* ingested by the sink, which depends on pollution emission rate, source-sink separation distance, and the time duration for which the source emits pollutants and the sink remains exposed. Here, the mean refers to a double average over many repetitions of the experiment (ensemble average over many turbulent realizations) and over all possible source-sink locations (spatial average for homogenization). (ii) *The standard deviation of cumulative pollutant concentration* ingested by the sink, which will serve as a measure of how far the cumulative concentration in an individual scenario deviates from the mean. The standard deviation of cumulative pollutant concentration can be further separated into a contribution arising from turbulent fluctuations and a contribution arising from inhomegeneity. The normalized standard deviation will also depend on the source-sink separation distance and the time duration for which the source emits pollutants and the sink remains

Journal Pre-proof

exposed.

In order to accurately estimate the mean and the standard deviation of cumulative pollutant concentration, (i) all possible turbulent flow scenarios must be sampled to account for the variability due to turbulence, (ii) all possible source-sink locations must be explored to account for the variability due to inhomogeneity, and (iii) we desire to decouple and quantify the above flow-dependent pollutant transport processes from the emission process. In the present work, we address these three challenges in the following way.

Variability due to turbulence: The straightforward construction of an ensemble of turbulent realizations requires many repetitions of the pollutant dispersal process. Under conditions of statistical stationarity, the expensive need for repeating the pollutant dispersal process numerous times can be avoided, and the ensemble average can be replaced with a time average. This assumes ergodicity and the turbulent dispersal process in each realization, over a long period of time, to explore all possible states of the true ensemble. In the present study, large eddy simulations of pollutant dispersal are carried out over a long time period. Uncorrelated temporal segments from this long simulation are then used as individual realizations for the ensemble average.

Variability due to inhomogeneity: The requirement to sample all possible source-sink combinations poses a significant challenge. In an experimental investigation or traditional CFD study, one must perform a new experiment or simulation for every source-sink combination within the room, thus requiring a very large number of experiments or simulations to gather the required statistics. Here, we avoid this with the novel *statistical overloading framework* whereby, with a single simulation, we obtain the cumulative exposure for every source-sink combination. This is achieved by considering a very large number of discrete sources spanning the entire region of interest and simultaneously emitting pollutant particles from all the sources. This way, one would have a large enough sample size of pollutant particles at any location for obtaining meaningful converged statistical averages. The region of interest will be similarly discretized into a very large number of sinks. As all the emitted particles are advected around by the turbulent flow, their position is carefully tracked to form the Lagrangian database. With post-processing, we can readily obtain the cumulative exposure for every source-sink combination. This simple strategy was made possible by the fact that the pollutant particles do not alter the flow (they are one-way coupled), and thus we can overload the turbulent flow with a significantly larger number of pollutant particles than in a typical scenario without altering the turbulent flow.

Decoupling of emission and transport processes: The pollutant concentration at the sink is the result of two processes: emission from the source followed by transport to the sink by the turbulent flow. Here, we decouple the two

Journal Pre-proof

processes and characterize the transport process in terms of delta function response (DFR) and step function response (SFR). The pollutant dispersal problem then reduces to accurate evaluation of the mean and two-time correlation of the DFR and SFR. The pollutant concentration at the sink in response to any emission process can then be expressed as a convolution with the DFR.

Based on the above discussion, this work will focus on the following three contributions:

- We present a rigorous statistical description of pollutant dispersal as a two-point, inhomogeneous, time-dependent turbulent process. Within this framework, we define the response to delta function emission and step function emission as fundamental quantities. We then derive relations that can be used to obtain the mean and standard deviation of cumulative pollutant concentration for any other emission process in terms of the fundamental delta function and step function responses (DFR and SFR).
- We then describe an Euler-Lagrange (EL) methodology for computing the DFR and SFR to any desired level of accuracy. The statistical overloading framework is described, which allows investigation of DFR and SFR for all possible source-sink combinations with a single simulation. Fundamental to this EL methodology is the ability to track the trajectory of millions of pollutant particles for an extended time period within the domain of interest, fully accounting for the effects of turbulence.
- The EL methodology requires careful selection of a number of computational parameters. These include the volumetric size of the discrete source and sink, the number of pollutant particles to be tracked from the source to the sink to obtain converged statistics, the number of turbulent realizations required for approximating the ensemble average, and the sampling frequency at the sink location. Considering the computational cost of Lagrangian tracking, statistically-informed guidelines will be developed to optimize computational parameters and achieve the desired level of accuracy and precision.

In this work, we demonstrate the above contributions in the context of indoor airborne disease transmission using data from EL large eddy simulations by Salinas et al. [22]. However, the Euler-Lagrange computational strategy that is described in this work, along with the statistical overloading technique, can be applied in any EL simulation to obtain the DFR and SFR under other turbulent pollution dispersal scenarios. Provided long time Lagrangian statistics of a large number of particles are obtained, the DFR and SFR can be accurately computed and then used to infer mean and standard deviation of cumulative pollutant concentration for any emission scenario. In the present large eddy simulation, we accurately account for the effect of unresolved sub-

Journal Pre-proof

grid turbulence on pollutant trajectories using a Langevin model. If the entire turbulence is resolved using direct numerical simulation, then there is no need for a subgrid model in computing the pollutant trajectories. Whereas in a Reynolds-averaged EL simulation, the effect of all the turbulent scales must be accounted for in computing the pollutant trajectories [33, 28]. Thus, the EL methodology outlined in this work can be used in conjunction with DNS, LES, and RANS simulations of pollutant dispersal.

The statistical overloading methodology has so far been primarily utilized to study dispersal in ventilated indoor spaces [22, 23, 21, 28, 33, 34]. Adapting this approach to other frameworks, including urban air quality management [35] and droplet impingement on airfoils [36] has yielded promising results. The article is organized as follows. In Section 2, we first define the problem of turbulent pollutant dispersal in a statistical framework for one chosen source-sink combination. Next, we express the mean and standard deviation of pollutant concentration at the sink in terms of the mean and two-time correlation of the DFR. We then describe the statistical overloading technique and proceed to discuss computational requirements to capture these DFR statistics to any desired level of accuracy and confidence in the context of a single source-to-sink combination. In Section 3, we extend the approach to consider all possible source-sink combinations to obtain rigorous estimations of the level of variation due to spatial inhomogeneity. This section also provides guidelines for the proper choice of additional computational parameters involved in this homogenization process. In Section 4, we discuss the results and summarize the main conclusions in Section 5.

2. Statistical Description

In this section, we discuss the quantities of interest (QoI) in quantifying dispersal of pollutants in a turbulent flow. The problem is stated as follows: given a time-dependent *source* of pollutant located at $\mathbf{x}_{so}$ emitting pollutant at the rate $q_{so}(t)$ for $t \geq 0$, we are interested in establishing the pollutant concentration $c\left(t, \mathbf{x}_{so}, \mathbf{x}_{si}\right)$ at a receiver or a sensor (henceforth referred to as the *sink*) located at $\mathbf{x}_{si}$. We immediately note that $c\left(t, \mathbf{x}_{so}, \mathbf{x}_{si}\right)$ is a stochastic quantity that varies from realization to realization. With the emission process and the source/sink locations being the same, the pollutant concentration measured in one run of the experiment will not be precisely the same as in another. The difference between the various realizations stem from stochastic variations arising from two factors: randomness in the emission process at the source and randomness in the transport process from the source to the sink.

Regarding the first factor, for example, in the case of virus-laden airborne droplet nuclei emitted by a sick individual or pollutant leaking from a punc-

Journal Pre-proof

tured container, the rate of emission expected in an individual incidence may not be precisely and deterministically known. In such cases where there is variability in the emission process, the emission process can only be characterized statistically. A detailed description of the stochastic emission process requires the M-time probability density function (PDF) (see Appendix A for details). Such complete statistical description is however, hard to achieve, and therefore, the statistics of pollutant emission is often simplified to the two leading order moments: the one-time statistics of mean emission, $\langle q_{so} \rangle(t)$ and the two-time correlation of emission fluctuation, $\langle q'_{so}(t_1)\, q'_{so}(t_2) \rangle$, where $q'_{so}(t) = q_{so}(t) - \langle q_{so} \rangle(t)$. The latter is often further simplified to obtain the standard deviation of emission as $\sigma_{q_{so}}(t) = \langle (q'_{so}(t))^2 \rangle^{1/2}$ (see Appendix A).

The second source of variability is due to fluctuations arising from the turbulent flow responsible for transporting the pollutant from the source to the sink. We quantify this turbulent transport process in terms of the *delta function response* (DFR), also known as the Green's function (see Appendix B for further details). The DFR is defined as the pollutant concentration $c_\delta(t, \mathbf{x}_{so}, \mathbf{x}_{si})$ measured at the sink at time t after a unit emission at the source at time zero. For simplicity, we assume the turbulent flow to be statistically stationary. As a result, DFR depends only on the time elapsed since emission, with the time of emission being statistically irrelevant.

For any source-sink combination, the time evolution of the DFR will vary from one realization to the next, depending on the nature of turbulent transport in that realization. Thus, c_δ is a random function that depends on the turbulent nature of pollutant transport, which is independent of the emission process. Of particular interest to the present work are the first two moments: the one-time ensemble-averaged DFR, $\langle c_\delta \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si})$, that can be calculated by averaging over an ensemble of realizations and the two-time correlation of the DFR fluctuation, $\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle$, where $c'_\delta(t) = c_\delta(t, \mathbf{x}_{so}, \mathbf{x}_{si}) - \langle c_\delta \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si})$. Again, the latter can be used to calculate the standard deviation of the DFR as $\sigma_{c_\delta}(t, \mathbf{x}_{so}, \mathbf{x}_{si}) = \langle (c'_\delta(t))^2 \rangle^{1/2}$ (see Appendix C).

2.1. Concentration in terms of the Delta Function Response

We now consider the pollutant concentration $c(t)$ at the sink to be a product of the above two stochastic processes - i.e., pollutants randomly ejected over time from the source are subsequently randomly transported by the prevailing turbulent flow to the sink. The probability distribution of $c(t)$ can be constructed in terms of the emission and transport (DFR) probability distributions (see Appendix C). In this work, we will focus on a limited set of leading order statistics of the PDF, namely the mean and the standard deviation of pollutant concentration. These represent the two key factors essential for accurately estimating the risk of exposure to the pollutant. Although higher-order

statistical quantities (skewness, flatness, two-time correlations, etc.) are not considered in this work, they can easily be formulated following the steps to be pursued below. We denote the mean concentration as $\langle c\rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si})$ and the standard deviation of pollutant concentration at the sink as

$$\langle\sigma_{c,1}\rangle^2(t, \mathbf{x}_{so}, \mathbf{x}_{si}) = \langle (c'(t, \mathbf{x}_{so}, \mathbf{x}_{si}))^2\rangle = \langle c^2\rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}) - \langle c\rangle^2(t, \mathbf{x}_{so}, \mathbf{x}_{si})\,, \quad (1)$$

where $c'(t, \mathbf{x}_{so}, \mathbf{x}_{si}) = c(t, \mathbf{x}_{so}, \mathbf{x}_{si}) - \langle c\rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si})$. We define $\langle\sigma_{c,1}\rangle$ with a single angle bracket to imply that it is an ensemble-averaged quantity and the subscript "1" implies that the variation is due to emission and turbulent transport (later we will use subscript "2" to imply variation is due to spatial inhomogeneity). In the above, we explicitly include the dependence on both the source and the sink locations. Although not shown, we can include dependence on other parameters, such as the size (radius) of the dispersing pollutant R, which can vary from sub-micron-sized aerosols to larger particles and droplets.

In Appendix C, we rigorously obtain expressions for these statistical quantities in terms of the emission and the DFR statistics. The mean pollutant concentration at the sink can be written in terms of the ensemble-averaged mean DFR-concentration $\langle c_\delta\rangle$ and the mean emission $\langle q_{so}\rangle$ as

$$\langle c\rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}) = \int_0^t \langle q_{so}\rangle(t_1)\, \underbrace{\langle c_\delta\rangle(t - t_1, \mathbf{x}_{so}, \mathbf{x}_{si})}_{\substack{\text{mean DFR}\\ \text{concentration}}}\, dt_1\,. \quad (2)$$

The integral represents the Duhamel's superposition theorem, where the mean concentration at time t depends on the convolution between the mean emission at time t_1 and the mean turbulent transport over the time interval $t - t_1$. The linear superposition uses the fact that the pollutants are one-way coupled and do not alter the turbulent flow.

The corresponding expression for the standard deviation is (see Appendix C)

$$\begin{aligned}\langle\sigma_{c,1}\rangle^2(t, \mathbf{x}_{so}, \mathbf{x}_{si}) = &\int_0^t\int_0^t \langle q_{so}\rangle(t - t_1)\,\langle q_{so}\rangle(t - t_2)\quad \underbrace{\langle c'_\delta(t_1)\, c'_\delta(t_2)\rangle}_{\substack{\text{2-time transport}\\ \text{fluctuation correlation}}}\quad dt_1\, dt_2 + \\ &\int_0^t\int_0^t \underbrace{\langle q'_{so}(t_1) q'_{so}(t_2)\rangle}_{\substack{\text{2-time emission}\\ \text{fluctuation correlation}}}\ \langle c_\delta\rangle(t - t_1)\,\langle c_\delta\rangle(t - t_2)\, dt_1\, dt_2\,,\end{aligned} \quad (3)$$

where $c'_\delta = c_\delta - \langle c_\delta\rangle$ is the concentration fluctuation in a realization of the DFR. For ease of interpretation, compared to the appendix, we have (i) replaced ζ' by q'_{so} and ξ' by c'_δ and (ii) swapped (t_1, t_2) with $(t - t_1, t - t_2)$ in the

Journal Pre-proof

Journal Pre-proof

first convolution integral. Thus, $\sigma_{c,1}$ arises from both turbulent fluctuations acting on the mean emission (first term on the right-hand side) and from the randomness of the emission process transported by the mean transport process (second term on the right-hand side). Due to their uncorrelatedness, the combined randomness of emission and transport has been ignored, as discussed in Appendix C.

In essence, for accurate evaluation of the mean and standard deviation of pollutant concentration, we require the mean and the two-time correlation of the emission and the turbulent transport processes. Furthermore, henceforth, we will consider the emission process to be an externally specified input and thus focus only on the effect of turbulent transport. In other words, in what follows, we will discuss methodologies for accurate evaluation of the required statistical properties of the DFR, namely $\langle c_\delta \rangle$ and $\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle$.

2.2. Accurate Evaluation of Delta Function Response in Euler-Lagrange Simulations

In this section, we discuss a Lagrangian Monte Carlo approach to computing the discretized delta function response. In a CFD simulation with Lagrangian tracking of particles, consider a representative source region of volume V_s around the location $\mathbf{x}_{so}$. We randomly release $\mathcal{N}_{so}$ particles within this source volume at $t = 0$ over a short duration of Δt. The particles originating from the source are monitored over the duration of the simulation to see how many of them are observed within a representative volume of V_s around the sink location $\mathbf{x}_{si}$ at some later time t after the initial release. Here, t is the time each individual particle takes to reach the sink from its initial release at the source location.

Let $\mathcal{N}_{si}(t)$ be the number of particles that were observed within the sink volume. The ratio $c_\delta(t) \approx \mathcal{N}_{si}(t)/(\mathcal{N}_{so} V_s)$ can be interpreted as the pollutant concentration at time t at the sink in response to a discrete delta function emission at $t = 0$ at the source, i.e., the DFR of an individual realization for a particular choice of source and sink.

As an example, we present results from the large eddy simulation (LES) of indoor pollutant dispersal in a $10 \times 10 \times 3.2$ m^3 room with a 4-way cassette air-conditioner unit installed on the ceiling at the center of the room [22, 23] as shown in Figure 1. The ventilation system is comprised of four inlets of size 0.055m $\times$ 0.44m surrounding an outlet of size 0.6m $\times$ 0.6m (see Fig. 1C). The air conditioner unit operates at an ACH of 2.5 with perfect filtration, i.e., none of the pollutants exiting the room through the outlet are recirculated back into the room through the AC inlets.

The gas-phase governing equations are the incompressible Navier–Stokes equations. The filtered gas-phase governing equations have been rigorously

Journal Pre-proof

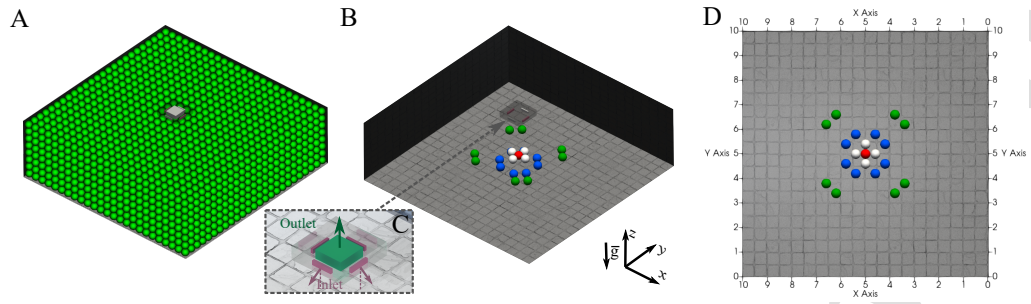


Figure 1: (A) Schematic of the canonical room, including 5000 spherical sub-volumes that can operate as sources or sinks for obtaining the Lagrangian database. (B) 4-way cassette air conditioning unit is placed at the center of the room. (C) Positions of the spherical subvolumes with the source represented by a red sphere at the center of the room and the sinks distributed at separation distances of 0.4 m (4 white spheres), 1 m (8 blue spheres), and 2 m (8 green spheres). These locations are illustrated in (C) an isometric view and (D) the top view.

derived in the context of Euler-Lagrange multiphase flow simulations [37, 38]. In the present limit of very low pollutant volume fraction, the governing equations can be further simplified to obtain the following equations

$$\nabla \cdot \tilde{\mathbf{u}} = 0\,, \tag{4}$$

$$\frac{\partial \tilde{\mathbf{u}}}{\partial t} + \tilde{\mathbf{u}} \cdot \nabla \tilde{\mathbf{u}} = -\nabla \tilde{p} + (\nu + \nu_t)\, \nabla^2 \tilde{\mathbf{u}}\,, \tag{5}$$

where tilde indicates the filtered variable. The sub-grid Reynolds stress term has been closed with the eddy viscosity model, where the turbulent eddy viscosity ν_t is obtained using the dynamic Smagorinsky model [39, 40].

As for the Lagrangian phase, each pollutant particle is individually tracked. The governing equations for the position and velocity of the lth pollutant are given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\mathbf{X}_l \\ \mathbf{V}_l\end{bmatrix} = \begin{bmatrix}\mathbf{V}_l \\ \mathbf{F}_l/m_l\end{bmatrix}. \tag{6}$$

Here, m_l is the mass of the lth nucleus. For the contribution from the unresolved subgrid eddies to the perturbation velocity of the particles, we utilize a Langevin model [41, 42, 43, 44, 45]. For pollutants that are smaller than 10μm an Equilibrium-Lagrangian model [46] is used.

The gas-phase LES equations are solved using a highly scalable spectral element solver, Nek5000 [47]. The Lagrangian droplets are solved using the highly scalable point-particle library ppiclF [48]. The domain is discretized using $60 \times 60 \times 16$ hexahedral elements (polynomial order of five) with 6^3

Journal Pre-proof

Gauss-Lobatto-Legendre (GLL) grid points within each element. This results in a total of 12.4 million grid points. 2 million pollutant particles each of radii 0.2 and 1 μm, are randomly but uniformly introduced within the room. The nuclei are then transported by the flow, where they would eventually settle to the floor, deposit on the walls and ceiling, or be removed by the outlet. The cost of each simulation amounts to approximately 221,000 CPU hours. For additional details on the numerical methodology, the reader is referred to the detailed descriptions in previously published works [22, 23].

The source $\mathbf{x}_{so}$, depicted as a red sphere in Figure 1, is positioned at the center of the room, equidistant from all four walls and elevated 1 m above the floor. The mean velocity magnitudes within the room and within the source location are observed to be 0.0746 m/s and 0.0432 m/s, respectively. The sink $\mathbf{x}_{si}$ is located at the same elevation of 1 m but at a distance r from the source. This source-sink combination, along with the air-conditioner within the room, is shown in Figure 1. The discrete delta function response from 50 different realizations is shown in Figures 2A, 2B, and 2C for three different separation distances of $r = 0.4$, 1 and 2 m, respectively.

All the DFRs display the same qualitative behavior. There is an initial delay period during which c_δ remains zero, since it takes a finite time for the pollutant to advect to the sink and correspondingly the delay increases with r. Following the initial delay, c_δ sharply increases to reach a peak and then slowly decays back to zero. These DFRs were calculated based on realizations containing on average $\langle \mathcal{N}_{so} \rangle \approx 13000$ particles emitted at the source. A fraction of these particles, as evidenced by c_δ, are later observed at the sink, where the sampling frequency was set at 1 s intervals. Figure 2A indicates that in the different realizations, a maximum of about 0.5% to 5% of the emitted $\mathcal{N}_{so}$ particles are observed within the sink volume located at a distance $r = 0.4$ m from the source, at around 5 s after emission. As the separation distance r increases, fewer particles are observed at the sink, as indicated by the reduced DFR peak. On the other hand, realization-to-realization fluctuations increase as a result of the reduced number of particles observed at the sink. For all separation distances, we observe substantial realization-to-realization variation due to the chaotic nature of the turbulent transport. For instance, in some scenarios, the flow at the moment of emission is directed from the source toward the sink, leading to faster arrival and higher pollutant concentrations at the sink. Conversely, in other scenarios, the flow moves away from the sink, causing slower arrival and lower concentrations.

For every source-sink combination, in general, we have considered an ensemble of $\mathcal{R} = 160$ realizations, from which the approximation of $\langle c_\delta \rangle$ is obtained by averaging over these realizations. The ensemble average for the particular source-sink combinations shown in Figures 2A, 2B, and 2C are pre-

Journal Pre-proof

sented in panels D, E, and F, respectively. It must be emphasized that the fine-grained discrete DFR of individual realizations shown in panels A, B, and C is dependent on the choice of the discretization parameter V_s. In contrast, the ensemble-averaged $\langle c_\delta \rangle$ is more robust and less sensitive to the choice of V_s.

We tried different well-known distributions such as exponential, beta, gamma, and binomial distributions to fit the DFRs presented in Figure 2. We observe the time evolutions of the DFR, including the initial delay period, peak and the later slow decay, to be best approximated by the following modified version of the Birnbaum-Saunders (or Fatigue life) distribution [49, 50]:

$$c_{\delta,l}(t, \mathbf{x}_{so}, \mathbf{x}_{si}) = \frac{A_l}{2\mu_l^2 \gamma_l^2 \sqrt{\pi}} \left(\frac{t^2 - \mu_l^2}{\sqrt{\frac{t}{\mu_l}} - \sqrt{\frac{\mu_l}{t}}} \right) \exp\left[\frac{1}{\gamma_l^2} \left(\frac{t}{\mu_l} + \frac{\mu_l}{t} - 2 \right) \right], \quad (7)$$

where γ is the shape parameter, μ is the scale parameter, and A is the long-time integrated DFR exposure. The subscript l denotes the lth realization.

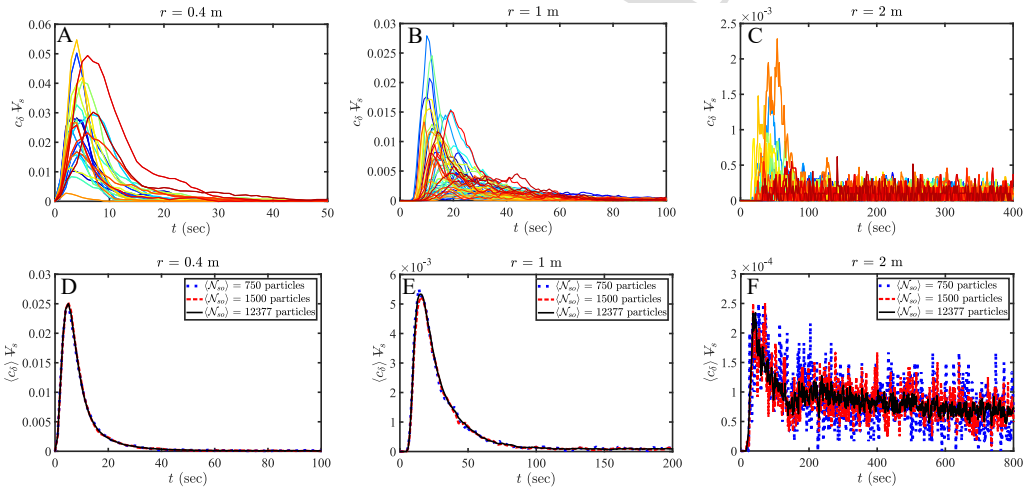


Figure 2: $c_\delta V_s$ vs time for 50 out of the 160 available realizations for separation distances of (A) $r = 0.4\,\text{m}$, (B) $r = 1\,\text{m}$, and (C) $r = 2\,\text{m}$. In all realizations, the source is positioned at the center of the room (i.e. $x_{so} = y_{so} = 5$ m) at an elevation of 1 m from the floor (i.e. $z_{so} = 1$ m). The room dimensions are $10 \times 10 \times 3.2\,\text{m}^3$. The bottom panels show the ensemble averages, $\langle c_\delta \rangle V_s$ vs time for different sample sizes, i.e. values of $\mathcal{N}_{so}$, to showcase the impact of a reduced sample size. $\mathcal{N}_{so} = 12377$ (black lines), $\mathcal{N}_{so} = 1500$ (red dashed lines), and $\mathcal{N}_{so} = 750$ (blue dotted lines) for (D) $r = 0.4$, (E) $r = 1$ and (F) 2 m. All quantities are computed over 160 turbulent realizations.

The values of A_l, μ_l, and γ_l were evaluated for the different realizations for each source-sink combinations of a certain separation distance r. Fig-

Journal Pre-proof

ure 3 shows the normalized histograms of $(A_l - \langle A \rangle)/\sigma_A$, $(\mu_l - \langle \mu \rangle)/\sigma_\mu$, and $(\gamma_l - \langle \gamma \rangle)/\sigma_\gamma$ for $r = 1$ and 2 m, respectively. Here the mean and standard deviation of A, μ, and γ are evaluated for each r and denoted as $\{\langle A_l \rangle, \sigma_A\}$, $\{\langle \mu_l \rangle, \sigma_\mu\}$, and $\{\langle \gamma_l \rangle, \sigma_\gamma\}$. Based on the histograms, we conclude that A, μ, and γ reasonably follow a normal distribution.

The two-time correlation of the DFR $\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle$ is plotted in Panels A to F of Figure 4 for the specific source-sink combination considered in Figure 2 for $r = 0.4$, 1, and 2 m respectively. The correlation is expected to be large when the two times are the same, i.e., when $t_1 = t_2$. Furthermore, we expect the correlation to be zero for $t_1, t_2 < t_{crit}$, where t_{crit} is the minimum time required for the pollutant to be transported from the source to the sink. The correlation increases and reaches a peak at a time delay of $t_1 = t_2 = t_{max}$. The value of this peak time depends on the source-sink separation distance r. As t_1 and t_2 increase beyond the peak time, the correlation falls back to zero due to the decay of the DFR. From the figure, we observe $t_{max} = 5$, 12, and 45 s for $r = 0.4$, 1, and 2 m. Also, at these distances, the correlation decays back to zero in approximately 20, 30, and 80 s.

The increased fluctuations seen in the case of $r = 2$ m in Figure 2 due to limited sample size also contribute to the estimated variance of pollutant concentration. This sample size-dependent contribution is over and above that due to turbulent fluctuations and the latter is the focus of this study. We can filter out the sample size-dependent contribution by subtracting its influence on variance by modeling it as a shot noise Poisson point process [51].

2.3. Statistical Overloading Technique

In this section, we describe the *statistical overloading technique* that allows evaluation of an ensemble of DFRs for *all* possible source-sink combinations within the room. Such complete information is then used in Section 3 to obtain homogenized results through spatial averaging. In the present context of the indoor space shown in Figure 1, the room is divided into N_V equal sub-volumes, each with a spherical shape and volume V_s. For any source-sink configuration, any one of the N_V sub-volumes can act as a source or sink of pollutants. If we exclude the possibility of the source and sink being collocated, there are $N_V(N_V - 1)$ source-sink combinations within the room.

With the room being divided into thousands of sub-volumes, we face the challenge of computing the DFR of millions of source-sink combinations. The statistical overloading technique is a convenient approach to obtaining these exhaustive statistics in an efficient manner. In this approach, millions of particles are initially distributed throughout the room and subsequently tracked over a long duration. As particles leave the room, either through the ventilation outlet or through deposition on the walls or the floor, new particles

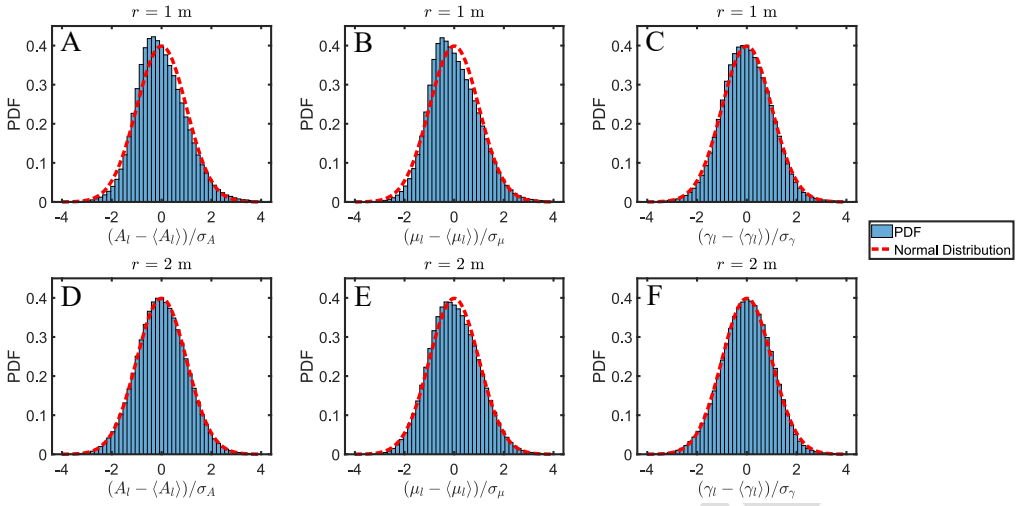


Figure 3: The probability density function of the standardized parameters of the modified Birnbaum-Saunders distribution (See Eqn. (7)) for all source-sink combinations available at (A,B,C) $r = 1$ m and (D,E,F) $r = 2$ m. Each source-sink combination has 160 realizations of $c_{\delta,l}$ and the corresponding fit parameters, A_l, μ_l and γ_l are standardized by subtracting the mean and dividing by the standard deviation. For comparison, the dashed red lines indicate the probability density function of a standard normal distribution.

can be randomly reintroduced into the room to maintain a constant average concentration [1].

The time history of all the computed particles forms the Lagrangian database used in constructing the DFR statistics. The Lagrangian database is used to identify all the particles that traveled from within the source sub-volume to the sink sub-volume. The times t_{so} and t_{si} when the particles were observed at the source and sink, respectively, are noted. All the particles with their source observation time within a short time window of Δt are then considered to form a realization of the discrete delta function emission. By sorting particles according to their travel time (i.e., $t_{si} - t_{so}$), we calculate the delta function response $c_{\delta,l}(t_{si} - t_{so}, \mathbf{x}_{so}, \mathbf{x}_{si})$ of an individual realization. Note that the individual realization does not constitute a continuous release of duration Δt. Rather, travel time data for all particles collected over the release window Δt are averaged to estimate the delta function response by assuming that Δt is smaller than the large eddy timescales of the flow. In this way, the travel times are assumed to be sampled from the same underlying delta function response.

[1] Although maintaining a constant overall concentration of particles is not essential, it simplifies the construction of the DFR statistics.

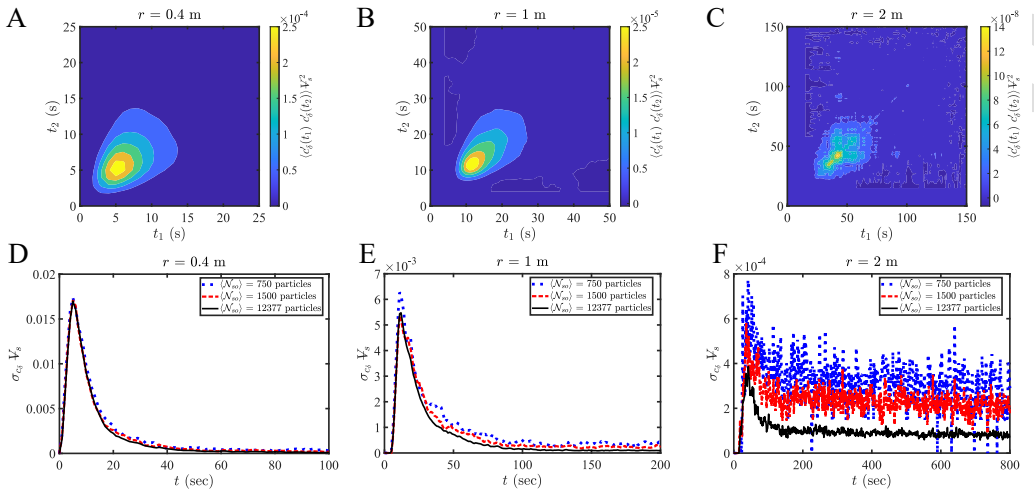


Figure 4: Colored contour plots (D, E, F) show $\langle c'\delta(t_1), c'\delta(t_2)\rangle$ over 160 realizations for separation distances of $r = 0.4$ m, 1 m, and 2 m. The temporal evolution of the standard deviation due to turbulent fluctuations, $\langle\sigma_{c,1}\rangle$ (black lines), is shown for separation distances (D) $r = 0.4$ m, (E) $r = 1$ m, and (F) $r = 2$ m. Also plotted are $\langle\sigma_{c,1}\rangle$ for (D) $r = 0.4$ m, (E) $r = 1$ m, and (F) $r = 2$ m, where the average number of particles per realization, $\mathcal{N}_{so}$, is reduced to 1500 (red dashed lines) and 750 (blue dotted lines), illustrating the impact of reduced sample size. All quantities are obtained over 160 turbulent realizations.

We refer to this approach as the statistical overloading technique for the following reasons. Instead of computing the ensemble of the DFR for each source-sink combination with millions of independent simulations, here by performing a single simulation where we track a large number of pollutant particles we compute the DFR of all possible source-sink combinations. Furthermore, a single simulation can be used to construct an ensemble of $\mathcal{R}$ realizations by considering different reasonably separated departure time windows of size Δt. The turbulent flow transporting the pollutant from the source to the sink differs in these time windows, forming independent realizations.

Physical scenarios involving dispersal of pollutants or aerosols, such as aerosol transmission of diseases, occur at the limit of a finite number of particles. The small sample size in such scenarios could result in observations that are different from what one expects the true population value to be. Statistical overloading also allows us to bypass this sampling variability in such scenarios. In the statistical overloading approach, we will consider the limit of copious emission of particles so that sampling variability will not be an issue. Scenarios where a small number of particles are emitted will be statistically overloaded by computing the scenario as if the emission occurs with a large number of particles. This allows one to focus attention on variability arising from turbulent fluctuations and spatial inhomogeneity. The sampling variability can be

Journal Pre-proof

easily estimated and treated as an added source of uncertainty.

2.4. Parameters of statistical overloading and their determination

Here, we discuss the choice of five computational parameters that are important in the implementation of the statistical overloading technique in an Euler-Lagrange simulation. These parameters must be carefully chosen since they affect the accuracy of computed $\langle c_\delta \rangle$ and $\langle c'_\delta(t_1)c'_\delta(t_2)\rangle$ and also determine the computational cost. They are as follows:

1. The total number of pollutant particles per realization released at the source and monitored as they travel to the sink must be sufficiently large. In general, the requirement for accurate calculation of the second-order statistics (two-time correlation) is more demanding than that for the mean.
2. The number of turbulent realizations $\mathcal{R}$ used in calculating the ensemble average. Here again, the requirement for accurate calculation of the second order statistics is more demanding than that for the mean.
3. The size of the source and sink volumes within which the pollutant particles are located.
4. The sampling window Δt used to ensemble average the delta function over a single turbulent realization.
5. The sampling frequency δt at which to gather the Lagrangian statistics.

These five parameters play a key role and determine the adequacy and accuracy of the Lagrangian pollutant data to be gathered from the Euler-Lagrange simulation. The criteria for choosing these parameters apply for every source-sink combination. Although the discussion will be based on the present problem of indoor pollutant dispersal, the results to be obtained can be generalized to other pollutant dispersal simulations.

Total number of particles to be emitted for accurate estimation of the mean: In the definition $c_\delta = \mathcal{N}_{si}/(\mathcal{N}_{so}V_s)$, the number of particles $\mathcal{N}_{so}$ observed within the source volume V_s over the time Δt constitutes the ensemble averaged delta function release over a turbulent realization. We note that the fine-grain DFR $c_\delta(t)$ is computed based on the travel time distribution of the $\mathcal{N}_{so}$ particles in a single realization of a discrete delta function release. Whereas, the ensemble average $\langle c_\delta \rangle(t)$ is computed based on the travel time distribution of a much larger set of $\mathcal{R}\langle \mathcal{N}_{so}\rangle$ particles, where $\langle \mathcal{N}_{so}\rangle$ is the average number of particles emitted at the source in the ensemble of $\mathcal{R}$ realizations.

The problem of pollutant particles emitted from the source volume can be considered as $\mathcal{R}\langle \mathcal{N}_{so}\rangle$ identical independent trials. The particle being observed at the sink at any time t is treated as a successful event. The mean probability of success, $\tilde{p}$, can be defined as the average number of pollutants being observed

Journal Pre-proof

at the sink. This yields the definition

$$\tilde{p} = \frac{\mathcal{R}\,\langle \mathcal{N}_{si} \rangle}{\mathcal{R}\,\langle \mathcal{N}_{so} \rangle} = \langle c_\delta \rangle\, V_s\,. \tag{8}$$

From Figure 2 we observe that this probability of detecting the pollutant at the sink at any given later time is generally quite small. Nevertheless, by considering a sufficiently very large number of pollutant particles emitted from the source (i.e., by considering $\mathcal{R}\,\langle \mathcal{N}_{so} \rangle \gg 1$) we can make the number of pollutant particles observed at the sink, $\mathcal{R}\,\langle \mathcal{N}_{si} \rangle$, to be large. In this limit, the probability of success out of the total $\mathcal{R}\,\langle \mathcal{N}_{so} \rangle$ trials approaches the normal distribution. The validity of the normal distribution assumption is somewhat supported by the fact that the parameter A of the modified Birnbaum-Saunders distribution, which is proportional to the fraction of pollutants reaching the sink, is normally distributed (See Figure 3 A,D).

We now want estimate the minimum number of trials, i.e., the minimum value of $\mathcal{R}\,\langle \mathcal{N}_{so} \rangle$, required to calculate $\langle c_\delta \rangle\, V_s$ to an accuracy $100\,(1-\beta)\%$ with a confidence interval of $100\,(1-\alpha)\%$. Using properties of normal distribution, this requirement can be expressed as

$$\mathcal{R}\langle\, \mathcal{N}_{so} \rangle > \max_t \left\{ \frac{Z^2_{\alpha/2}}{\beta^2} \frac{1}{\tilde{p}} \frac{\sigma^2_{c_\delta}}{\langle c_\delta \rangle^2} \right\}, \tag{9}$$

where $Z_{\alpha/2}$ is the Z-score of the normal distribution corresponding to a cumulative distribution value of $(1-\alpha/2)$. In the above equation, both $\tilde{p}$ and the ratio of standard deviation to mean DFR, $\sigma_{c_\delta}/\langle c_\delta \rangle$, are time-varying quantities and thus the minimum requirement of $\mathcal{R}\,\langle \mathcal{N}_{so} \rangle$ must be based on the maximum value of the right hand side over the desired time interval. From Figures 2 and 4 we note that the ratio of standard deviation to mean value of DFR is typically less than unity and later we will observe this ratio to further decrease upon homogenization. Here we proceed with the approximation $\sigma_{c_\delta}/\langle c_\delta \rangle \approx 1/2$ to establish the minimum emission requirement. Furthermore, we desire to obtain $\langle c_\delta \rangle\, V_s$ values greater than 0.002 to 95% accuracy at 95% confidence. With these choices of $\alpha = \beta = 0.05$ ($Z_{\alpha/2} = 1.96$), and $\tilde{p} = 0.002$, we obtain $\mathcal{R}\langle \mathcal{N}_{so} \rangle > 192,040$. A plot of $\mathcal{R}\langle \mathcal{N}_{so} \rangle$ versus $\tilde{p}$ is shown in Figure 5A for $\alpha = \beta = 0.05$ for two different values of $\sigma_{c_\delta}/\langle c_\delta \rangle$. The two curves are off by a factor of four.

With $\mathcal{R} = 160$ realizations, this yields a requirement of $\langle \mathcal{N}_{so} \rangle \geq 1200$ to achieve this accuracy. This prediction is tested in Figures 2D-F where we plot $\langle c_\delta \rangle V_s$ for $r = 0.4$, 1 and 2 m. The black lines present well-converged results obtained with $\langle \mathcal{N}_{so} \rangle = 12377$, which corresponds to all the particles at the chosen source location in the Lagrangian database. We then calculated $\langle c_\delta \rangle V_s$

Journal Pre-proof

with randomly chosen sub-samples of only $\langle \mathcal{N}_{so} \rangle = 1500$ (red lines) and 750 (blue lines) particles. We observe that for the shorter separation distances, $r = 0.4$ and 1 m, there is little to no variation with the reduction in the sample size. A larger level of fluctuation is observed for $r = 2$ m. Since the value of $\tilde{p} = \langle c_\delta \rangle V_s$ is more than an order of magnitude lower, the level of accuracy and confidence suffers. This is also consistent with the fact that the level of fluctuation increases with sub-sampling.

Total number of particles to be emitted for accurate estimation of two-time correlation:

The inequality (9) can be modified to estimate the minimum value of $\mathcal{N}_{so}$ required for accurate evaluation of the fine-grain discrete DFR c_δ of an individual realization by setting $\mathcal{R} = 1$ on the left hand side. This much stricter requirement is not met even with the full Lagrangian dataset of $\langle \mathcal{N}_{so} \rangle = 12377$. Thus, we do not achieve an accuracy of $\beta = 0.05$ at 95% confidence. If we maintain the minimum value of accurate prediction to be $c_\delta V_s = 0.002$, we can reevaluate the actual accuracy achieved to be $\beta = 0.2$ or 0.4, depending on whether we choose the ratio $\sigma_{c_\delta}/\langle c_\delta \rangle = 0.5$ or 1.0. Thus, the fine grained DFRs are obtained only to 60 to 80% accuracy with 95% confidence. This is reflected in the plots presented in Figure 2 where the individual realizations are noisy, while the ensemble average is substantially smoother.

For accurate estimation of σ_{c_δ} we require both the individual fine grain DFRs to be accurate and have enough realizations to approximate the true ensemble. We will examine the latter requirement on $\mathcal{R}$ in a following subsection. The lower accuracy in the evaluation of fine grain $c_\delta(t)$ will propagate to the estimation of the two-time correlation and standard deviation estimation as well. The impact of this is examined in Figures 4G-I, where we plot $\langle \sigma_{c,1} \rangle V_s$ for $r = 0.4$, 1, and 2 m. The black lines show well-converged results obtained with $\langle \mathcal{N}_{so} \rangle = 12377$. The red and blue lines show $\langle \sigma_{c,1} \rangle V_s$ for randomly chosen sub-samples of $\langle \mathcal{N}_{so} \rangle = 1500$ and 750 particles. Good convergence is observed for $r = 0.4$ m and also for $r = 1$ m. At larger distances, increasing noise is observed. It can thus be concluded that even the two time correlation is evaluated with sufficient accuracy in the present simulations. Furthermore, it must be pointed out that in the following sections we will introduce homogenization where we average over many source-sink combinations and evaluate cumulative exposure where we integrate over time. Both these operations tend to increase the number of samples and thereby yield much higher accuracy for the second order statistics.

Number of realizations $\mathcal{R}$ required for accurate estimation of mean:

The inequality given in (9) is only a necessary condition for the accurate evaluation of $\langle c_\delta \rangle$. In addition, we require the $\mathcal{R}\langle \mathcal{N}_{so} \rangle$ Lagrangian particles to

Journal Pre-proof

be drawn from a sufficiently large ensemble of realizations, so that the DFR adequately samples all possible turbulent transport scenarios. We estimate the adequacy of $\mathcal{R}$ in the following way. As noted in the earlier section, the fine-grain $c_\delta(t)$ can be approximated by the modified Birnbaum-Saunders distribution, characterized by the amplitude A_l, scale parameter μ_l, and shape parameter γ_l, which are approximately normally distributed. The adequacy of $\mathcal{R}$ can be evaluated indirectly in terms of how well we sample the normal distributions of A, μ, and γ.

The accuracy of the sample mean, obtained with $\mathcal{R}$ realizations, in approximating the true mean can be evaluated using standard statistical tools. In the present context, the mean is being estimated along with the simultaneous estimation of the unknown variance. Hence, to evaluate the mean to within $100(1-\beta)\%$ accuracy with $100(1-\alpha)\%$ confidence the following conditions must be satisfied

$$\frac{\sigma_A}{\langle A\rangle}\frac{t_{\alpha/2,\mathcal{R}-1}}{\sqrt{\mathcal{R}}}\,,\ \frac{\sigma_\mu}{\langle\mu\rangle}\frac{t_{\alpha/2,\mathcal{R}-1}}{\sqrt{\mathcal{R}}}\,,\ \frac{\sigma_\gamma}{\langle\gamma\rangle}\frac{t_{\alpha/2,\mathcal{R}-1}}{\sqrt{\mathcal{R}}} < \beta\,, \tag{10}$$

which can be rearranged to obtain the criterion

$$\sqrt{\mathcal{R}} > \max\left[\frac{\sigma_A}{\langle A\rangle},\ \frac{\sigma_\mu}{\langle\mu\rangle},\ \frac{\sigma_\gamma}{\langle\gamma\rangle}\right]\frac{t_{\alpha/2,\mathcal{R}-1}}{\beta}\,, \tag{11}$$

where $t_{\alpha/2,\mathcal{R}-1}$ is the student t-distribution value at $\alpha/2$ for $\mathcal{R}-1$ degrees of freedom. The three terms on the left-hand side correspond to estimating the mean value of the three parameters of the Birnbaum-Saunders distribution.

In the indoor pollution problem being considered, the β obtained from Eq. (10) is plotted as a function of $\mathcal{R}$ in Figure 5 for the two values of $r = 1$ and 2 m. For $\mathcal{R} = 160$, we obtain $\beta = 0.058$ and 0.04 for $r = 1$ and 2 m, respectively. This indicates that 95% accuracy with 95% confidence is exceeded with 160 realizations at $r = 2$ m, while at the shorter separation distance of $r = 1$ m, an ensemble of more than 216 realizations are needed to achieve 95% accuracy. The higher accuracy at $r = 2$ m is due to the lower ratio of the standard deviation to the mean of the Birnbaum-Sanders distribution parameters.

Panel A in Figure 6 plots the ensemble averaged concentration, $\langle c_\delta\rangle V_s$ for $r = 1$ m by successively reducing the number of realizations by randomly subsampling among the 160 realizations. In all the cases, we satisfy the earlier requirement of $\mathcal{R}\langle\mathcal{N}_{so}\rangle > 192,040$, and thereby, here we isolate and investigate the adequacy of ensemble size. We observe the convergence to be incomplete and exhibit a non-monotonic behavior due to the limited number of realizations considered. Convergence with increasing $\mathcal{R}$ will become clearer later when we consider a step function response.

The results plotted in Figures 5 and 6 are for the particular source-sink combinations shown in Figures 1B and C. We now investigate the condition given in Eq. (10) for all source-sink combinations of separation distance r in the room. For this, the inequality (10) can be restated as: in order to achieve $\alpha = 0.05$ and $\beta = 0.05$ with $\mathcal{R} = 160$ realizations, we require $\sigma_A/\langle A\rangle, \sigma_\mu/\langle\mu\rangle, \sigma_\gamma/\langle\gamma\rangle < 0.32$. In other words, we require the standard deviation to be less than one-third of the mean for all three parameters. Figure 7 shows normalized histograms of $\sigma_A/\langle A\rangle, \sigma_\mu/\langle\mu\rangle, \sigma_\gamma/\langle\gamma\rangle$ obtained for all the source-sink combinations at separation distances of $r = 1$ m (A, B, C) and $r = 2$ m (D, E, F). We achieve our requirement of $\alpha = 0.05$ and $\beta = 0.05$ for nearly two-thirds of the source-sink combinations at $r = 1$ m and more than 92 % of the source-sink combinations at $r = 2$ m.

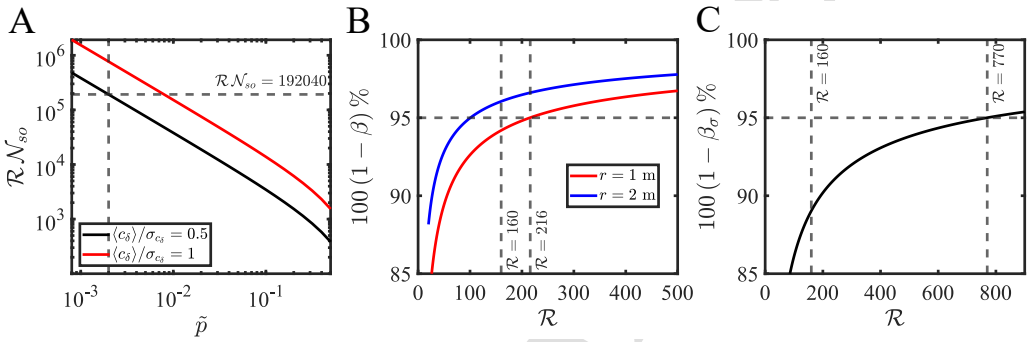


Figure 5: (A) Minimum number of particles required originating from the source as a function of $\tilde{p}$ according to Eq. (9) for two values of $\langle c_\delta\rangle/\sigma_{c_\delta}$. Accuracies of the (B) mean concentration and (C) standard deviation due to turbulent fluctuations as a function of number of realizations $\mathcal{R}$ using Eqs. (11) and (12) respectively. The conditions for the mean were obtained using the coefficients of the modified Birnbaum-Strauss distribution for the source and sinks ($r = 1$ and 2 m) shown in Figure 1 B and C. The values of the coefficients A, μ, and γ obtained by fitting for the 160 DFR realizations were used to obtain $\max\left[\langle A\rangle/\sigma_A\,,\ \langle\mu\rangle/\sigma_\mu\,,\ \langle\gamma\rangle/\sigma_\gamma\right]$ to be used in the LHS of Eq. (11).

Number of realizations $\mathcal{R}$ required for accurate estimation of two-time correlation:

We now proceed to estimate the accuracy of the computed correlation $\langle c'_\delta(t_1)c'_\delta(t_2)\rangle$ in the following way. With the mean values of the Birnbaum-Sanders distribution ($\langle A\rangle$, $\langle\mu\rangle$, and $\langle\gamma\rangle$) estimated, the accuracy of $\langle c'_\delta(t_1)c'_\delta(t_2)\rangle$ depends on the estimation of σ_A, σ_μ, and σ_γ. For a normal distribution, the ratio of true variance σ^2_{tr} to the computed variance can be bounded as

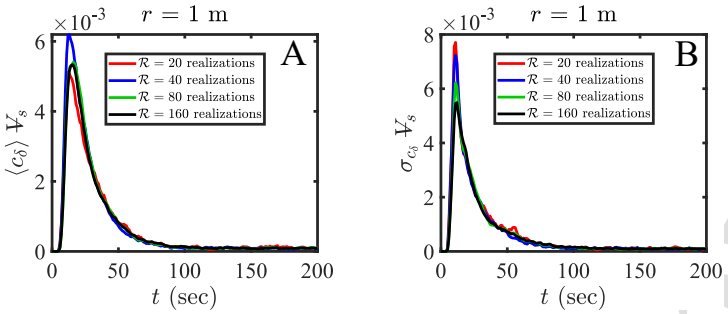


Figure 6: Normalized values of the ensemble averaged $\langle c_\delta \rangle$ and σ_{c_δ} for a separation distance of $r = 1$ m. The number of realizations is randomly reduced to 80, 40, and 20 realizations from the 160 available realizations.

$$\frac{\chi_1^2}{\mathcal{R}-1} \leq \left\{ \frac{\sigma_A^2}{\sigma_{A,tr}^2}, \frac{\sigma_\mu^2}{\sigma_{\mu,tr}^2}, \frac{\sigma_\gamma^2}{\sigma_{\gamma,tr}^2} \right\} \leq \frac{\chi_2^2}{\mathcal{R}-1}, \tag{12}$$

where χ_1^2 and χ_2^2 are χ^2-distributions of $(\mathcal{R}-1)$ degrees of freedom with cumulative distribution values $\alpha/2$ and $1-\alpha/2$, respectively. In the limit $\mathcal{R} \gtrapprox 100$, they can be approximated as $\chi_1^2 = (\sqrt{2(\mathcal{R}-1)} - Z_{\alpha/2})^2/2$ and $\chi_2^2 = (\sqrt{2(\mathcal{R}-1)} + Z_{\alpha/2})^2/2$, in terms of the Z-score of the normal distribution. In the present context of $\mathcal{R} = 160$ and $\alpha = 0.05$, we obtain the result, $0.814 \leq \sigma_{tr}^2/\sigma^2 \leq 1.27$.

Defining the accuracy of the calculated standard deviation as $\beta_\sigma = |\sigma - \sigma_{tr}|/\sigma_{tr}$, we obtain the following relation, $\beta_\sigma = 1 - \sqrt{\chi_1^2/(R-1)}$. In the present context of $\mathcal{R} = 160$ and $\alpha = 0.05$, we obtain an accuracy of 79.2%. Figure 5b shows the relation between β_σ and $\mathcal{R}$, where it can be inferred that nearly 770 realizations are required to capture the standard deviation to an accuracy of 95%.

The accuracy of the computed correlation $\langle c'_\delta(t_1) c'_\delta(t_2) \rangle$ depends on the accuracy of the Birnbaum-Sanders parameters, but their relation is complex. We evaluate the adequacy of $\mathcal{R}$ in the calculation of the two-time correlation by plotting in Panel B of Figure 6 the convergence of $\langle \sigma_{c,1} \rangle / q_{so}$ with a successive increase in the number of realizations. As expected, the uncertainty in the prediction of the standard deviation is significant. We do not observe any significant pattern in the convergence of standard deviation, which may be expected from the theoretical predictions of Eq. (12). Thus, uncertainties associated with the two-time correlation are substantially larger than those associated with the mean. Far more realizations are required to further reduce the uncertainty in the standard deviation and the two-time correlation.

Proper choice of the source and sink volume V_s:

Journal Pre-proof

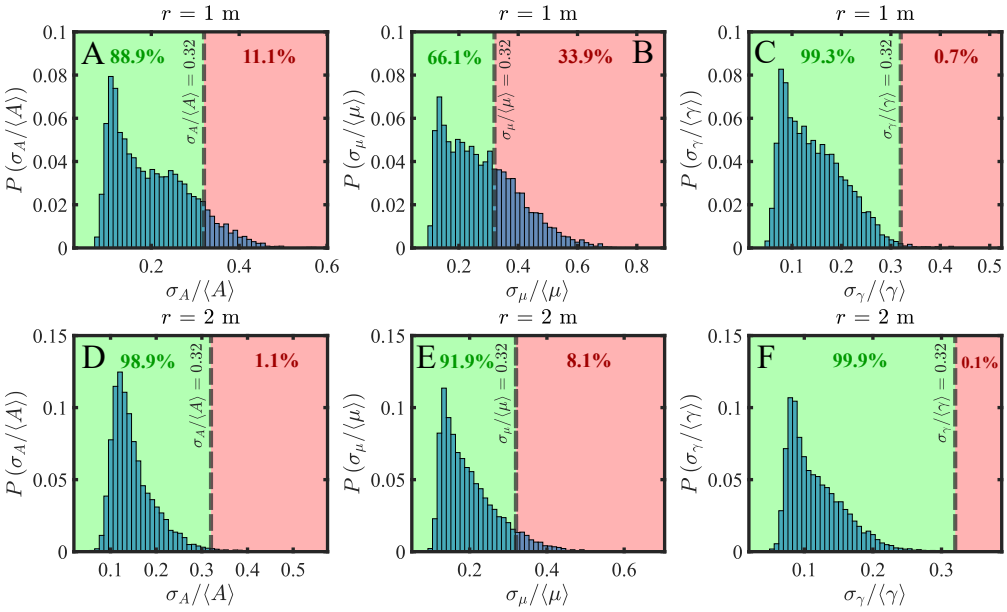


Figure 7: Probability density functions of the ratio of the standard deviation to the mean over 160 realizations of the modified Birnbaum-Saunders distribution for all source-sink combinations. The PDFs are plotted at separation distances of (B,C,D) $r = 1$ and (E,F,G) $r = 2$ m. The vertical dashed line is used to designate our required condition for $\alpha = 0.05$ and $\beta = 0.05$. The green shaded region indicates all the source-sink combinations that satisfy our required condition, and the red region indicates the ones which do not.

The evaluation of the discrete DFR depends on the definition of the source/sink volume V_s. In the present work, the room volume was divided into $25 \times 25 \times 8 = 5000$ source/sink sub-volumes. By decreasing V_s we can increase the number of sources and sinks within the room and thereby achieve higher resolution of the inhomogeneity within the room. However, with a smaller V_s, to achieve the minimum value of $\mathcal{R}\langle \mathcal{N}_{so} \rangle$ given in Eq. (9) and Section (2.4), either the number of realizations ($\mathcal{R}$) or the total number of particles seeded within the room must be proportionately increased. Another consideration when choosing V_s is the separation distance between the source and the sink. This is to ensure that there is no overlap between the source & sink volumes and make sure that edge effects don't distort our statistics. Therefore, for an accurate description of the effect of separation distance, the length scale of the source/sink volume ($\approx V_s^{1/3}$) must be smaller than the smallest separation distance. With the present choice of 5000 sub-volumes and 160 realizations, we obtain the estimate that the room must be seeded in excess of one million particles to achieve 95% accuracy at 95% confidence. With smaller source/sink volumes, we will need to track a correspondingly larger number of pollutant

Journal Pre-proof

particles within the room in order to achieve the same level of accuracy.

Proper choice of sampling window Δt and sampling frequency δt:

In the present work, we define the sampling window Δt to be the time over which we observe the particles that are within the source volume, where they constitute the total number of emitted particles, $\mathcal{N}_{so}$. The results presented above correspond to an sampling window of 200 s, where the ensemble-averaged outcomes appear to be insensitive to this choice. If Δt is decreased to smaller values, then the number of particles that make up a discrete delta function emission decreases, and correspondingly, the convergence or accuracy of the ensemble-averaged statistics degrades. On the other hand, a value of Δt much larger than the turbulent time scale is not appropriate, since it does not provide an accurate representation of an individual delta function realization. A larger value of Δt has the effect of averaging over many turbulent realizations. Thus, the estimation of $\langle c_\delta \rangle$ is not affected by the larger value of Δt (in fact, the number of realizations can be decreased with the longer Δt itself serving as multiple realizations). However, the statistics of fluctuations, such as $\langle c'_\delta(t_1) c'_\delta(t_2) \rangle$ will be under estimated with larger values of Δt.

The sampling frequency δt is the frequency at which the location of the pollutant particles within the room are noted in the simulation to create the Lagrangian database. In the present work, we choose $\delta t = 1$ s, and accordingly the time evolution of the DFR in Figure 2 and the two-time correlations in Figure 4 are plotted at 1 s resolution. From the smoothness of these plots, it is clear that a finer time resolution is not necessary. On the other hand, a much larger δt of the order of minutes will not yield meaningful plots. The proper choice of sampling frequency δt depends on the minimum source-sink distance and the maximum advective velocity that transports the pollutant particles.

We note that the minimum source-sink separation distance may be represented by the cubic root of the source/sink volume, which is given by $(V_s)^{1/3}$. This yields the requirement $\delta t < (V_s)^{1/3}/u_{max}$, where u_{max} is the maximum likely flow velocity advecting the pollutants. This criterion will guarantee that as a particle moves through a source/sink volume, its crossing will be recorded by its presence within the sink volume at one or more time instances. With a larger value of δt that violates the above criterion, there is a risk of the particle moving too quickly across the source/sink volume without being noted as it moved through the source/sink volume. In the current simulation of the canonical room with the ventilation system operating at ACH 2.5, more than 98% of the source volumes satisfy the above criterion for $\delta t = 1$ s. Upon closer inspection, we note that only the few source volumes that are very close to the inlet and outlet vents violate the criterion due to much stronger local fluid velocities. Fortunately, these sub-volumes are not important since they typically

Journal Pre-proof

are not occupied by a source or sink. Therefore, a sampling frequency of 1 s is sufficiently small enough to capture the required statistics without significant error.

3. Homogenization and Quantification of Spatial Variation

In most applications, including the present case of a ventilated indoor space, the problem of turbulent dispersal is spatially inhomogeneous. As a result, the mean and standard deviation of concentration depend on both $\mathbf{x}_{so}$ and $\mathbf{x}_{si}$. This dependence complicates the evaluation and interpretation of the results. *Homogenization*, which can be carried out at different levels, is often used to reduce the complexity of the problem. The most aggressive homogenization is to average over both $\mathbf{x}_{so}$ and $\mathbf{x}_{si}$. Such homogenization in the context of indoor pollutant dispersal gives rise to the well-mixed concentration that is independent of the source and sink locations [22, 30]. Averaging over only the source or sink locations leaves the mean and standard deviation dependent on either $\mathbf{x}_{so}$ or $\mathbf{x}_{si}$ [22, 34, 28]. Another simple homogenization strategy that is appropriate in many dispersal problems is to consider the source-sink distance $r = |\mathbf{x}_{so} - \mathbf{x}_{si}|$ as the only parameter. This allows averaging over all source-sink combinations that are separated by the distance r. Other homogenization strategies may be suitable for other dispersal problems.

When averaged over all possible source-sink combinations of a fixed separation distance, the mean concentration can be expressed as

$$\langle\langle c \rangle\rangle(t, r) = \frac{1}{S(r)} \iint_{|\mathbf{x}_{so} - \mathbf{x}_{si}| = r} \langle c \rangle(t_1, \mathbf{x}_{so}, \mathbf{x}_{si})\, d\mathbf{x}_{so}\, d\mathbf{x}_{si}\,, \tag{13}$$

where the normalization factor in the denominator is $S(r) = \iint_{|\mathbf{x}_{so} - \mathbf{x}_{si}| = r} d\mathbf{x}_{so}\, d\mathbf{x}_{si}$. The nature of this homogenization or averaging process must be clarified. In the above, we assume the emission process to be the same at all the source locations being averaged. The double angle bracket on the left-hand side implies an ensemble average over turbulent realizations followed by an average over all source-sink locations of distance r.

With this homogenization, the pollutant concentration at the sink in any individual realization will deviate from the double average concentration, $\langle\langle c \rangle\rangle$. This deviation from the double-average in an individual realization can be

Journal Pre-proof

quantified as follows

$$\begin{aligned}\langle\!\langle\sigma_c\rangle\!\rangle^2(t,r) &= \frac{1}{S(r)}\iint_{|\mathbf{x}_{so}-\mathbf{x}_{si}|=r}\langle(c-\langle\!\langle c\rangle\!\rangle)^2\rangle(t,\mathbf{x}_{so},\mathbf{x}_{si})\,d\mathbf{x}_{so}\,d\mathbf{x}_{si}\\ &= \langle\!\langle(c-\langle\!\langle c\rangle\!\rangle)^2\rangle\!\rangle = \langle\!\langle c^2\rangle\!\rangle - \langle\!\langle c\rangle\!\rangle^2\\ &= \underbrace{\langle\!\langle\langle c^2\rangle - \langle c\rangle^2\rangle\!\rangle}_{\langle\!\langle\sigma_{c,1}\rangle\!\rangle(t,r)} + \underbrace{\langle\!\langle\langle c\rangle^2\rangle - \langle\!\langle c\rangle\!\rangle^2}_{\langle\!\langle\sigma_{c,2}\rangle\!\rangle(t,r)}\,.\end{aligned} \tag{14}$$

The outer angle bracket corresponds to the spatial average and the inner angle bracket corresponds to the ensemble average. The overall variance $\langle\!\langle\sigma_c\rangle\!\rangle^2$ arises from two different contributions. The realization-to-realization variation due to turbulence was defined earlier in Eq. (1). It depends on the location of the source and the sink. This, when averaged over all source-sink separations of distance r, yields

$$\langle\!\langle\sigma_{c,1}\rangle\!\rangle^2(t,r) = \frac{1}{S(r)}\iint_{|\mathbf{x}_{so}-\mathbf{x}_{si}|=r}\langle c'^2\rangle(t,\mathbf{x}_{so},\mathbf{x}_{si})\,d\mathbf{x}_{so}\,d\mathbf{x}_{si}\,, \tag{15}$$

where $c' = c - \langle c\rangle$ is the concentration fluctuation from the ensemble average for a particular source-sink combination. The second contribution is due to $\langle c\rangle(t,\mathbf{x}_{so},\mathbf{x}_{si})$ at the different source-sink combinations deviating from the homogenized average $\langle\!\langle c\rangle\!\rangle(t,r)$. The standard deviation of this spatial variation can be evaluated as

$$\langle\!\langle\sigma_{c,2}\rangle\!\rangle^2(t,r) = \frac{1}{S(r)}\iint_{|\mathbf{x}_{so}-\mathbf{x}_{si}|=r}\left(\langle c\rangle(t,\mathbf{x}_{so},\mathbf{x}_{si}) - \langle\!\langle c\rangle\!\rangle(t,r)\right)^2\,d\mathbf{x}_{so}\,d\mathbf{x}_{si}\,. \tag{16}$$

In summary, in a pollutant dispersal problem, given the location of the source and the sink, the typical QoI are the ensemble-averaged mean and standard deviation, $\langle c\rangle(t,\mathbf{x}_{so},\mathbf{x}_{si})$ and $\langle\sigma_{c,1}\rangle(t,\mathbf{x}_{so},\mathbf{x}_{si})$. In problems where the source and sink locations are variables, the QoI can be simplified through homogenization. When the ensemble averages are averaged over all source-sink locations of a separation distance r, the resulting homogenized mean is $\langle\!\langle c\rangle\!\rangle(t,r)$. The standard deviation derives contribution both from realization-to-realization variation, $\langle\!\langle\sigma_{c,1}\rangle\!\rangle(t,r)$, and spatial variation in the ensemble average $\langle\!\langle\sigma_{c,2}\rangle\!\rangle(t,r)$.

3.1. In terms of the Homogenized Delta Function Response

In this section, we express the homogenized mean concentration and the standard deviations in terms of suitable statistics obtained from the delta

function response. The homogenized mean concentration given in Eq. (13) can be expressed as

$$\langle\!\langle c \rangle\!\rangle(t,r) = \int_0^t \langle q_{so} \rangle(t_1) \underbrace{\langle\!\langle c_\delta \rangle\!\rangle(t-t_1,r)}_{\substack{\text{homogenized mean} \\ \text{DFR concentration}}} dt_1 \,. \tag{17}$$

This convolution integral is analogous to Eq. (2), but for the homogenized quantity. In the spatial integration over all possible source-sink combinations separated by distance r, it is assumed that the mean emission $\langle q_{so} \rangle(t_1)$ remains the same. Thus, the homogenized pollutant concentration depends on the mean emission at time t_1 and homogenized mean transport over the time interval $t - t_1$, integrated over $0 \leq t_1 \leq t$.

The homogenized standard deviation of pollutant concentration due to realization-to-realization turbulent fluctuations can be expressed as

$$\begin{aligned} \langle\!\langle \sigma_{c,1} \rangle\!\rangle^2(t,r) = &\int_0^t \int_0^t \langle q_{so} \rangle(t-t_1) \langle q_{so} \rangle(t-t_2) \underbrace{\langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle\!\rangle}_{\substack{\text{homogenized 2-time} \\ \text{transport correlation}}} dt_1\, dt_2 + \\ &\int_0^t \int_0^t \langle q'_{so}(t_1) q'_{so}(t_2) \rangle \langle c_\delta \rangle(t-t_1) \langle c_\delta \rangle(t-t_2)\, dt_1\, dt_2 \,, \end{aligned} \tag{18}$$

where $\langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle\!\rangle$ is the two-time correlation of concentration fluctuation in a DFR, further averaged over all source-sink combinations of separation distance r. The second term that arises from the randomness of the emission process is taken to remain the same at all possible release locations. In this work, we ignore the second contribution due to our focus on understanding the role of turbulent dispersion.

The standard deviation of pollutant concentration due to spatial variation can be expressed as

$$\langle\!\langle \sigma_{c,2} \rangle\!\rangle^2(t,r) = \int_0^t \int_0^t \langle q_{so} \rangle(t-t_1) \langle q_{so} \rangle(t-t_2) \underbrace{\langle \langle c_\delta \rangle'(t_1) \langle c_\delta \rangle'(t_2) \rangle}_{\substack{\text{2-time correlation} \\ \text{of spatial variation}}} dt_1\, dt_2 \,, \tag{19}$$

where $\langle c_\delta \rangle' = \langle c_\delta \rangle - \langle\!\langle c_\delta \rangle\!\rangle$ is the spatial variation in the ensemble-averaged DFR from the homogenized concentration. Thus, $\langle \langle c_\delta \rangle'(t_1) \langle c_\delta \rangle'(t_2) \rangle$ corresponds to the two-time correlation of spatial fluctuation in the DFR.

Journal Pre-proof

In summary, we require the following delta function response statistics:

$$\begin{aligned}
\langle c_\delta \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}) &: \quad \text{unhomogenized mean concentration} \\
\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle(\mathbf{x}_{so}, \mathbf{x}_{si}) &: \quad \text{unhomogenized two-time correlation} \\
\langle\!\langle c_\delta \rangle\!\rangle(t, r) &: \quad \text{homogenized mean concentration} \\
\langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle\!\rangle(r) &: \quad \text{homogenized two-time correlation} \\
\langle \langle c_\delta \rangle'(t_1)\, \langle c_\delta \rangle'(t_2) \rangle(r) &: \quad \text{two-time correlation of spatial variation}\,.
\end{aligned} \tag{20}$$

From the first two, using Eqs. (2) and (3), we can compute the unhomogenized mean and standard deviation for any emission process. From the latter three, using Eqs. (17), (18), and (19), we can compute the homogenized mean and standard deviation for any emission process.

3.2. Step Function Response

For the homogenized results, we will present the outcomes of the step function response (SFR) analysis, as opposed to the previously considered DFR. This will be in alignment with prior models of indoor pollution, such as the well-mixed model and distance-dependent variants [30, 22, 2, 3, 4, 5, 6, 7, 8], where pollutant concentration was quantified as a function of time following the sudden initiation of a steady emission process at $t = 0$. The SFR is the pollutant concentration at the sink in response to a steady unit emission that was initiated at the source at $t = 0$. The step function emission corresponds to $q_{so} \equiv 1$ for $t \geq 0$ and as a result, $\langle q_{so} \rangle = 1$ and $\langle q'_{so}(t_1) q'_{so}(t_2) \rangle = 0$. The SFR can be calculated from the DFR by substituting these emission statistics as

$$\begin{aligned}
\langle n \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R) &= \int_0^t \langle c_\delta \rangle(t_1, \mathbf{x}_{so}, \mathbf{x}_{si}, R)\, dt_1\,, \\
\langle \sigma_{n,1} \rangle^2(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R) &= \int_0^t \int_0^t \langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle\, dt_1\, dt_2\,.
\end{aligned} \tag{21}$$

The step function responses for the specific source-sink combinations shown in Figure 1A are displayed in Figure 8. Panels A and B plot the ensemble-averaged concentration, $\langle n \rangle V_s$ for $r = 1$ and 2 m for four different values of $\mathcal{R}$. In all the cases, $\mathcal{R}\langle \mathcal{N}_{so} \rangle > 192,040$ and thereby we investigate convergence with increasing ensemble size. We note that the faster convergence predicted at a larger separation distance by the theory is observed in practice. The curves of mean concentration look almost identical even at values of $\mathcal{R} = 40$ for $r = 2$ m, while there is noticeable variation for $r = 1$ m. We evaluate the adequacy of $\mathcal{R}$ in the calculation of the two-time correlation by plotting in Panels C and D of Figure 6 show the convergence of $\langle \sigma_{c,1} \rangle / q_{so}$ for an increasing number of

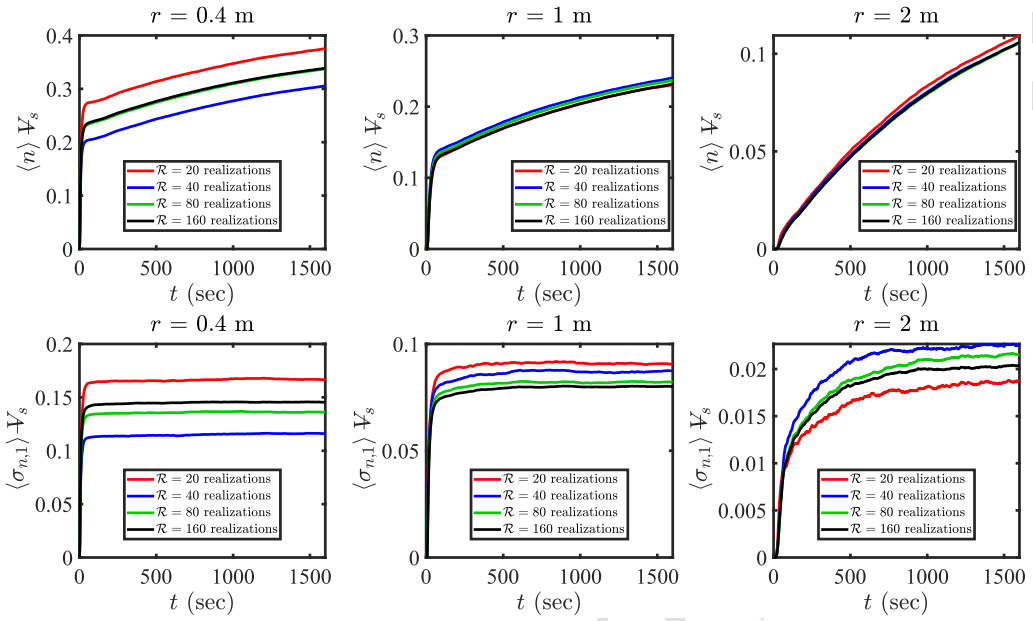


Figure 8: Normalized values of the step function response $\langle n \rangle$ and $\langle \sigma_{n,1} \rangle$ for separation distances of $r = 1$ and 2 m. The number of realizations is reduced to 80, 40, and 20 by sub-sampling out of the 160 available realizations.

realizations. As expected, the convergence in the prediction of the standard deviation is significantly slower than the mean. Unlike our observation of faster convergence at the larger separation distance in the case of the mean concentration, we do not observe any significant pattern in the convergence of standard deviation. We observe the standard deviation to be comparable to the mean at very early times of the order of a minute. Even at 1000 s, the standard deviation is about 40% of the mean at $r = 1$ m and 25% of the mean at $r = 2$ m. From the definition of standard deviation given in Eq. (21) and from the two-time correlation plots shown in Figure 4, it can be inferred that most of the fluctuation is due to randomness in the pollutant particles that arrive at the sink over the most recent 60 s or less.

3.3. Evaluation of Homogenized Results

In this section, ensemble-averaged DFR results are averaged over all possible source-sink locations of constant separation r. The homogenized mean

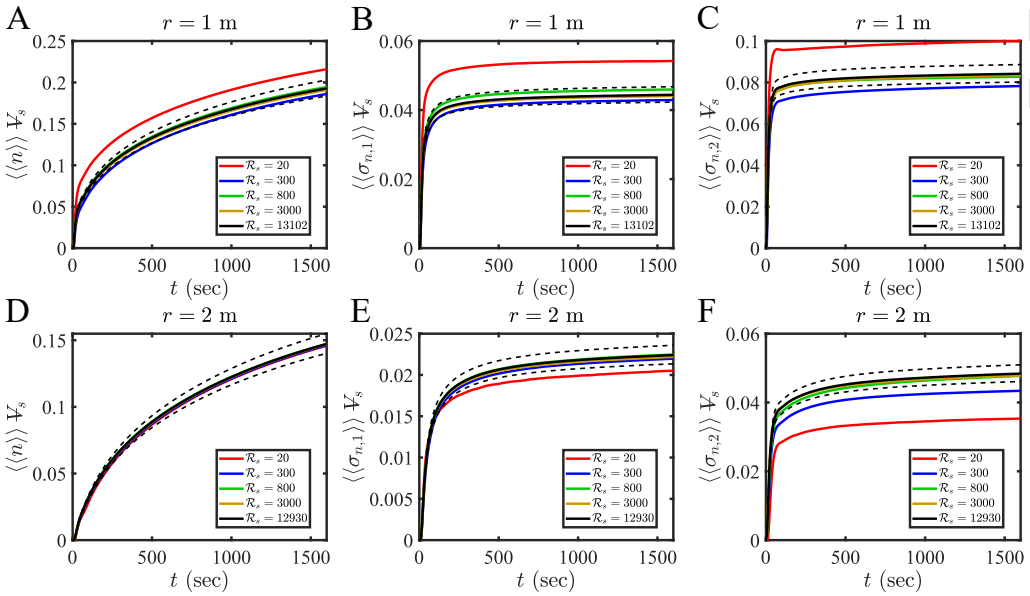


Figure 9: Temporal evolution of (A,D) $\langle\!\langle n \rangle\!\rangle$, (B,E) $\langle\!\langle \sigma_{n,1} \rangle\!\rangle$ and (C,F) $\langle\!\langle \sigma_{n,2} \rangle\!\rangle$ at separation distances $r = 1$ and 2 m. The black solid lines represent the quantities of interest at the maximum available value of $\mathcal{R}_s$ and the black dashed lines indicate the upper and lower bounds of 95% accuracy (i.e., $\beta = 0.05$). Random sub-sampling of the available realizations is used to approximate $\langle\!\langle n \rangle\!\rangle$, $\langle\!\langle \sigma_{n,1} \rangle\!\rangle$ and $\langle\!\langle \sigma_{n,2} \rangle\!\rangle$ at $\mathcal{R}_s = 20$, 300, 800, and 3000 to study the effect of decreasing the number of spatial realizations.

and standard deviations are defined as

$$\begin{aligned} \langle\!\langle n \rangle\!\rangle(t, r, R) &= \int_0^t \langle\!\langle c_\delta \rangle\!\rangle(t_1, r, R)\, dt_1 \\ \langle\!\langle \sigma_{n,1} \rangle\!\rangle^2(t, r, R) &= \int_0^t \int_0^t \langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle\!\rangle\, dt_1\, dt_2 \\ \langle\!\langle \sigma_{n,2} \rangle\!\rangle^2(t, r, R) &= \int_0^t \int_0^t \langle \langle c_\delta \rangle'(t_1)\, \langle c_\delta \rangle'(t_2) \rangle\, dt_1\, dt_2\,, \end{aligned} \tag{22}$$

The room was divided into $25 \times 25 \times 8$ sub-volumes of size 0.4 m along the three directions, which yields a total of 25 million source-sink combinations. These source-sink combinations were sorted into bins according to their center-to-center separation distance $r = |\mathbf{x}_{so} - \mathbf{x}_{si}|$. We denote the number of different source-sink combinations at a separation distance r to be $\mathcal{R}_s(r)$ which we dub as the number of spatial realizations. For example, there are 90,000 source-sink combinations at distances of $r = 1$ and 2 m. However, due to the square cross-section of the room on the horizontal plane, each source-sink combination in general has an eight-fold symmetry (there is only four-fold symmetry when both the source and sink fall on a diagonal or bisector vertical

Journal Pre-proof

plane). These symmetry combinations have been used in the calculation of the ensemble average. Therefore, approximately $(1/8)^{th}$ of the total source-sink combinations can be considered to capture the spatial variation (i.e., $\mathcal{R}_s \approx 13,102$).

The time evolution of the homogenized mean SFR at $r = 1$ and 2 m are shown in Panels A and D of Figure 9. The SFR increases steadily following the onset of emission at $t = 0$. In the present problem of the indoor space ventilated at ACH = 2.5, the time scale of the exponential increase is 1440 s. Accordingly, we see that the SFRs at $r = 1$ and 2 m are still evolving towards their steady state values. Here we focus on the black lines obtained using the entire dataset of $\mathcal{R}_s = 13,102$. As can be expected, the pollutant concentration is higher at the shorter distance of $r = 1$ m than at $r = 2$ m. The dependence of steady state concentration as a function of r has been studied in detail by [22], who observed the concentration to be higher than the room average for distances smaller than 4 m and lower at separation distances larger than 4 m. They also observed the time evolutions of the SFR to closely follow the exponential trend predicted by the well-mixed model.

Figure 10 plots the homogenized two-time correlation $\langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2)\rangle\!\rangle$ and the two-time correlation of spatial variation $\langle\langle c_\delta\rangle'(t_1)\, \langle c_\delta\rangle'(t_2)\rangle$ for separation distances $r = 1$ and 2 m. Similar to what was observed in Figure 4 for the non-homogenized two-time correlations, the homogenized correlations have a peak ridge along $t_1 = t_2$. Along this ridge, $\langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2)\rangle\!\rangle$ and $\langle\langle c_\delta\rangle'(t_1)\, \langle c_\delta\rangle'(t_2)\rangle$ reach their peak values at $t_{max} \approx 10$ and 17 for $r = 1$ and $r = 2$ m, respectively. The correlations subsequently decay to zero for $t_1 = t_2 > t_{crit}$, with $t_{crit} \approx 20$ and 40 s at the two separation distances. The homogenized two-time correlations are significantly smoother than those corresponding to specific source-sink locations, owing to the availability of a substantially larger dataset for averaging. Another observation worth noting is that the two-time correlation due to spatial variation has a marginally higher peak.

The standard deviations of the SFR, $\langle\!\langle\sigma_{n,1}\rangle\!\rangle$ and $\langle\!\langle\sigma_{n,2}\rangle\!\rangle$, are examined in Figure 9. First and foremost, we observe the total variation, measured as $\langle\!\langle\sigma_n\rangle\!\rangle = (\langle\!\langle\sigma_{n,2}\rangle\!\rangle^2 + \langle\!\langle\sigma_{n,2}\rangle\!\rangle^2)^{1/2}$, to be substantial compared to the mean. Furthermore, the standard deviations approach their steady state much faster in about 600 s. As a result, the ratio $\langle\!\langle\sigma_n\rangle\!\rangle/\langle\!\langle n\rangle\!\rangle$ is large at early times and decreases as t increases. For example, pollutant concentration at the sink in response to a step function release at the source in an individual realization can deviate from the estimated mean by almost 90% at early times of $t \lessapprox 500$ s. As $t \to \infty$, we observe the ratio $\langle\!\langle\sigma_n\rangle\!\rangle/\langle\!\langle n\rangle\!\rangle$ to be 0.48 and 0.35 at $r = 1$ and 2 m, respectively.

In Figure 9, we observe $\langle\!\langle\sigma_{n,2}\rangle\!\rangle$ to be nearly twice the value of $\langle\!\langle\sigma_{n,1}\rangle\!\rangle$. This highlights the fact that more of the variation in the pollutant concentration

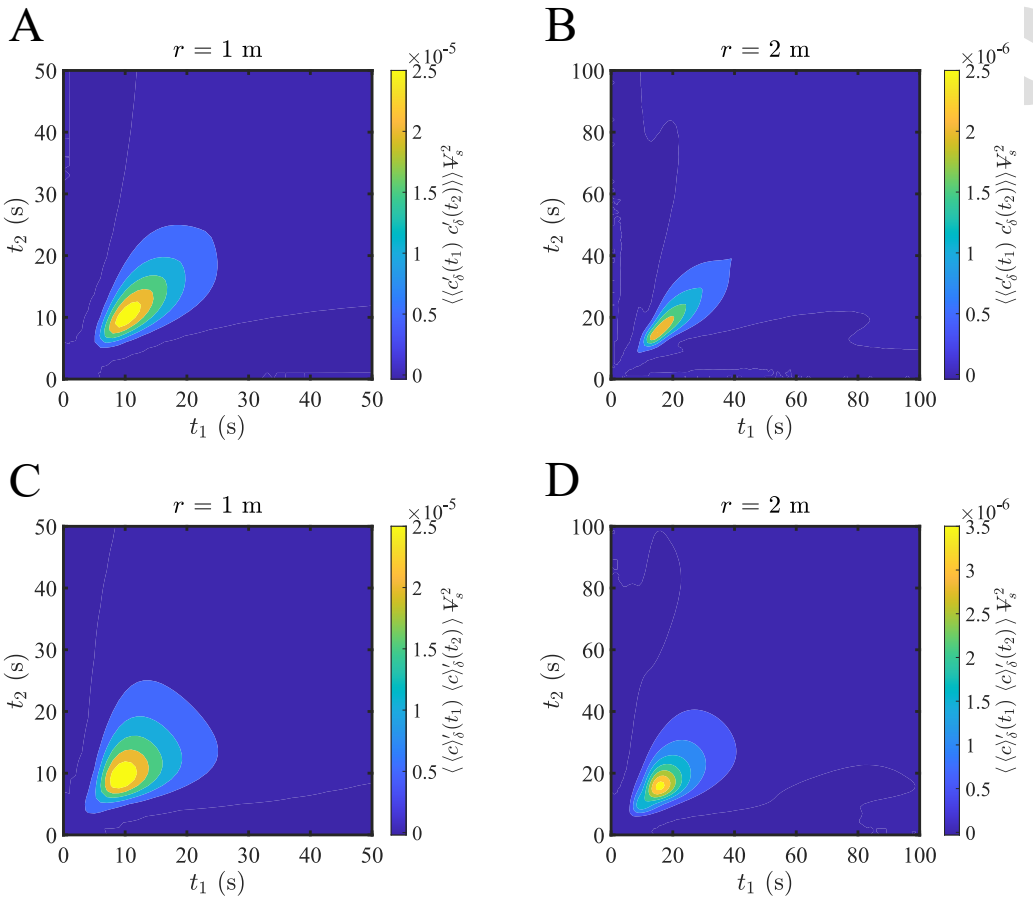


Figure 10: Contour plots of (A,B) homogenized two-time correlation and (C,D) two-time correlation due to spatial variation for $r = 1$ and 2 m.

is due to the variation in the spatial locations of the source-sink combination. Although realization-to-realization variation due the turbulent flow is non-negligible, exposure to the pollutants at the sink can be minimized by carefully choosing the source-sink locations.

Proper choice of $\mathcal{R}_s$:

To estimate the accuracy of homogenized statistics being evaluated, we must first determine the nature of the distribution. In the present context, we are interested in the distribution of ensemble-averaged DFRs for the different source-sink combinations of a certain separation distance. As presented in Figure 2, the ensemble-averaged DFR of each source-sink combination can be approximated by the Birnbaum-Sanders function. So the variation of ensemble averaged DFR for different source-sink combinations can be examined by look-

Journal Pre-proof

ing at the distribution of Birnbaum-Sanders parameters. These distributions were presented in Figure 7, where it is clear that they are not Gaussian. These distributions, due to spatial inhomogeneity, can be contrasted from those seen in Figure 3, which show a Gaussian distribution of Birnbaum-Sanders parameters due to turbulence.

Nevertheless, due to the large value of $\mathcal{R}_s$, the following condition can be taken to apply

$$\sqrt{\mathcal{R}_s} > \max\left[\frac{\sigma_A}{\langle A\rangle}, \frac{\sigma_\mu}{\langle\mu\rangle}, \frac{\sigma_\gamma}{\langle\gamma\rangle}\right] \frac{Z_{\alpha/2}}{\beta}\,. \tag{23}$$

In Eq. (11), the mean and standard deviation of the Birnbaum-Sanders parameters correspond to the fine-grain DFR calculated for various turbulence realizations at a particular source-sink location. Whereas, in Eq. (23), the mean and standard deviation of the Birnbaum-Sanders parameters are for the ensemble-averaged DFR calculated for various source-sink locations. Furthermore, for large $\mathcal{R}_s$, $t_{\alpha/2,R_s-1} \to Z_{\alpha/2}$. Using the values of $\sigma_A/\langle A\rangle$, $\sigma_\mu/\langle\mu\rangle$, and $\sigma_\gamma/\langle\gamma\rangle$ extracted from the distributions presented in Figure 7, we evaluate Eq. (23) to obtain the requirement of $\mathcal{R}_s > 326$ and $\mathcal{R}_s > 180$ for $r = 1$ and 2 m. These conditions strictly apply to the ensemble-averaged DFR, but here we will take them to apply to the mean SFR concentration as well.

Since $\langle\!\langle\sigma_{n,1}\rangle\!\rangle$ is the ensemble average of $\langle\sigma_{n,1}\rangle$ calculated for the $\mathcal{R}_s$ source-sink combinations, we evaluate the minimum number of samples required for calculating $\langle\!\langle\sigma_{n,1}\rangle\!\rangle$ to $100(1-\beta)\%$ accuracy with $100(1-\alpha)\%$ confidence as

$$\sqrt{\mathcal{R}_s} > \frac{\sigma_{\langle\sigma_{n,1}\rangle}}{\langle\!\langle\sigma_{n,1}\rangle\!\rangle} \frac{Z_{\alpha/2}}{\beta}\,, \tag{24}$$

where $\sigma_{\langle\sigma_{n,1}\rangle}$ is the standard deviation of $\langle\sigma_{n,1}\rangle$ computed for the $\mathcal{R}_s$ samples. Since $\langle\!\langle\sigma_{n,1}\rangle\!\rangle$ (see Figure 9) and $\sigma_{\langle\sigma_{n,1}\rangle}$ are time-dependent, here we evaluate the above criterion at the final steady state. We obtain the requirement $\mathcal{R}_s > 900$ and $\mathcal{R}_s > 665$ for $r = 1$ and 2 m, respectively.

Finally, we proceed to obtain the criterion for evaluating $\langle\!\langle\sigma_{n,2}\rangle\!\rangle$ accurately. Since this is the standard deviation of spatial variation we apply the criterion

$$\frac{\chi_1^2}{\mathcal{R}_s - 1} \le \frac{\langle\!\langle\sigma_{n,2}\rangle\!\rangle^2}{\langle\!\langle\sigma_{n,2}\rangle\!\rangle_{tr}^2} \le \frac{\chi_2^2}{\mathcal{R}_s - 1}\,. \tag{25}$$

For large $\mathcal{R}_s$, we earlier presented the approximations $\chi_1^2 = (\sqrt{2(\mathcal{R}-1)} - Z_{\alpha/2})^2/2$ and $\chi_2^2 = (\sqrt{2(\mathcal{R}-1)} + Z_{\alpha/2})^2/2$, in terms of the Z-score of the normal distribution. With the accuracy requirement of $|1 - (\langle\!\langle\sigma_{n,2}\rangle\!\rangle^2/\langle\!\langle\sigma_{n,2}\rangle\!\rangle_{tr}^2)| < \beta$ at $(100(1-\alpha)\%$ confidence, we obtain the requirement of $\mathcal{R}_s > 770$ for $\langle\!\langle\sigma_{n,2}\rangle\!\rangle$. This requirement should stand true independent of the choice of separation distance.

Journal Pre-proof

The vast sample size of spatial variation (i.e., large value of $\mathcal{R}_s$) allows us to test the validity of the three conditions that were presented above based on the assumption of normal distribution. To do that, we evaluate the confidence interval (i.e., α) numerically through a frequentist resampling approach for the quantities of interest as a function of $\mathcal{R}_s$ for the chosen value of $\beta = 0.05$. To determine the value of α achieved for any sample size $\mathcal{R}_s$, we randomly subsample $\mathcal{R}_s$ source-sink combinations out of the ∼13,000 available realizations and calculate $\langle\langle n \rangle\rangle_\sim$, $\langle\langle \sigma_{n,1} \rangle\rangle_\sim$ and $\langle\langle \sigma_{n,2} \rangle\rangle_\sim$ from the subspace. The subscript $\sim$ is introduced to denote that these quantities, computed with the smaller number of $\mathcal{R}_s$ samples, are less accurate than those computed with the complete dataset. The temporal evolution of $\langle\langle n \rangle\rangle_\sim$, $\langle\langle \sigma_{n,1} \rangle\rangle_\sim$ and $\langle\langle \sigma_{n,2} \rangle\rangle_\sim$ are plotted for different values of $\mathcal{R}_s$ in Figure 9.

We repeat this random resampling process 10,000 times and calculate that many values of $\langle\langle n \rangle\rangle_\sim$, $\langle\langle \sigma_{n,1} \rangle\rangle_\sim$ and $\langle\langle \sigma_{n,2} \rangle\rangle_\sim$. We then evaluate the fraction of the 10,000 repetitions in which the computed statistics fall within $100(1 \pm \beta)\%$ of the true statistics computed with all the ∼13,000 samples. This fraction is nothing but $(1 - \alpha)$. This process is repeatedly done for different values of $\mathcal{R}_s$. The results of this analysis are presented in Figure 11. For the mean SFR $\langle\langle n \rangle\rangle$, we achieve better than 95% confidence ($\alpha < 0.05$) for $\mathcal{R}_s = 283$ and $\mathcal{R}_s = 161$ at separation distances, $r = 1$ and 2 m, respectively. Similarly, for $\langle\langle \sigma_{n,1} \rangle\rangle$, we achieve 95% confidence for $\mathcal{R}_s = 864$ and $\mathcal{R}_s = 627$ at $r = 1$ and 2 m, respectively. These findings are similar to those obtained earlier with the assumption of normal distribution. However, we observe some difference in the estimation of $\langle\langle \sigma_{n,2} \rangle\rangle$. In order to achieve 95% accuracy ($\beta = 0.05$) at 95% confidence, we require a minimum $\mathcal{R}_s = 1819$ and 2725 at $r = 1$ and 2 m, respectively.

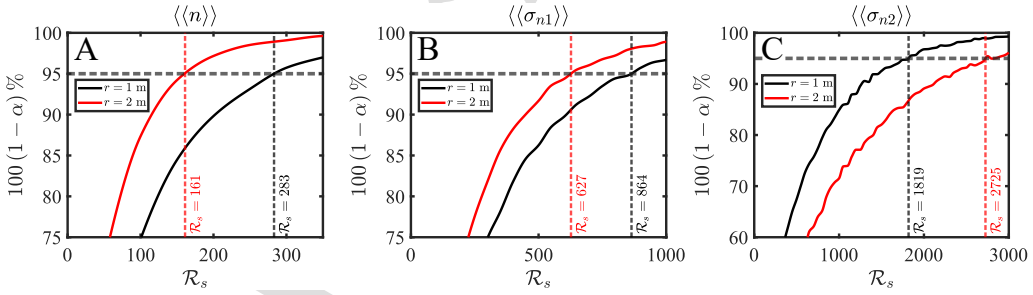


Figure 11: The confidence interval, α, vs $\mathcal{R}_s$ for (A) $\langle\langle n \rangle\rangle$, (B) $\langle\langle \sigma_{n,1} \rangle\rangle$ and (C) $\langle\langle \sigma_{n,2} \rangle\rangle$ for $r = 1$ and $r = 2$ m. The vertical dashed lines correspond to the minimum $\mathcal{R}_s$ required to achieve a 95 % confidence at $r = 1$ (black dashed lines) and $r = 2$ m (red dashed lines).

Journal Pre-proof

4. Discussion

4.1. Cumulative Exposure

So far, the focus has been on instantaneous pollutant concentration in response to discrete delta and step function emissions. However, the influence of the pollutant depends on the cumulative exposure of the sink over a period of time. If we consider the response to a step function release at the source location $\mathbf{x}_{so}$ starting from $t = 0$, the ensemble averaged mean cumulative pollutant concentration at the sink location $\mathbf{x}_{si}$ exposed over the time period $0 \leq t \leq T$ can be expressed as

$$\langle \overline{n} \rangle (T, \mathbf{x}_{so}, \mathbf{x}_{si}, R) = \int_0^T \langle n \rangle (\tau, \mathbf{x}_{so}, \mathbf{x}_{si}, R)\, d\tau = \int_0^T \left(\int_0^\tau \langle c_\delta \rangle (t_1, \mathbf{x}_{so}, \mathbf{x}_{si}, R)\, dt_1 \right) d\tau \,, \tag{26}$$

where the overbar indicates cumulation over the time period. The corresponding variance (square of the standard deviation) can be obtained in terms of the delta function fluctuation as

$$\langle \overline{\sigma_{n,1}} \rangle^2 (T, \mathbf{x}_{so}, \mathbf{x}_{si}, R) = \int_0^T \int_0^T \left(\int_0^{\tau_1} \int_0^{\tau_2} \langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle \, dt_1\, dt_2 \right) d\tau_1\, d\tau_2 \,. \tag{27}$$

The above source-sink-dependent mean and variance of cumulative pollutant concentration can be homogenized by further averaging over all possible source-sink combinations of separation r. The homogenized mean and variance of cumulative pollutant concentration can be expressed as

$$\begin{aligned}
\langle\!\langle \overline{n} \rangle\!\rangle (T, r, R) &= \int_0^T \left(\int_0^\tau \langle\!\langle c_\delta \rangle\!\rangle (t_1, r, R)\, dt_1 \right) d\tau \,, \\
\langle\!\langle \overline{\sigma_{n,1}} \rangle\!\rangle^2 (T, r, R) &= \int_0^T \int_0^T \left(\int_0^{\tau_1} \int_0^{\tau_2} \langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2) \rangle\!\rangle \, dt_1\, dt_2 \right) d\tau_1\, d\tau_2 \,, \\
\langle\!\langle \overline{\sigma_{n,2}} \rangle\!\rangle^2 (T, r, R) &= \int_0^T \int_0^T \left(\int_0^t \int_0^t \langle \langle c_\delta \rangle' (t_1)\, \langle c_\delta \rangle' (t_2) \rangle \, dt_1\, dt_2 \right) d\tau_1\, d\tau_2 \,.
\end{aligned} \tag{28}$$

From the above expressions it can be seen that the mean and variance of cumulative concentration (both unhomogenized and homogenized) can be related to time integrals of the step function response of pollutant concentrations discussed in the previous sections. Some important differences can be inferred. In Figure 9, the ratio $\langle\!\langle \sigma_{n,1} \rangle\!\rangle / \langle\!\langle n \rangle\!\rangle$ is observed to reach a steady state on the time scale of air circulation within the indoor space. This steady-state value represents the realization-to-realization variation in concentration expected at the sink for $t \gg 1/\mathrm{ACH}$. It also corresponds to the temporal fluctuations

Journal Pre-proof

observed at the sink within any single realization. In contrast, the corresponding ratio for cumulative exposure, $\langle\langle\overline{\sigma_{n,1}}\rangle\rangle/\langle\langle\overline{n}\rangle\rangle \to 0$, due to time integration over positive and negative fluctuations. On the other hand, since ensemble and time averages are interchangeable in stationary flows, it can be argued that the normalized standard deviation due to spatial inhomogeneity, given by $\langle\langle\overline{\sigma_{n,2}}\rangle\rangle/\langle\langle\overline{n}\rangle\rangle$, is minimally affected by cumulation and therefore closely resembles $\langle\langle\sigma_{n,2}\rangle\rangle/\langle\langle n\rangle\rangle$. In essence, for large values of T, variation in cumulative exposure is expected to be most influenced by spatial inhomogeneity.

4.2. Evaluation of Computational Parameters for Larger Pollutant Particles

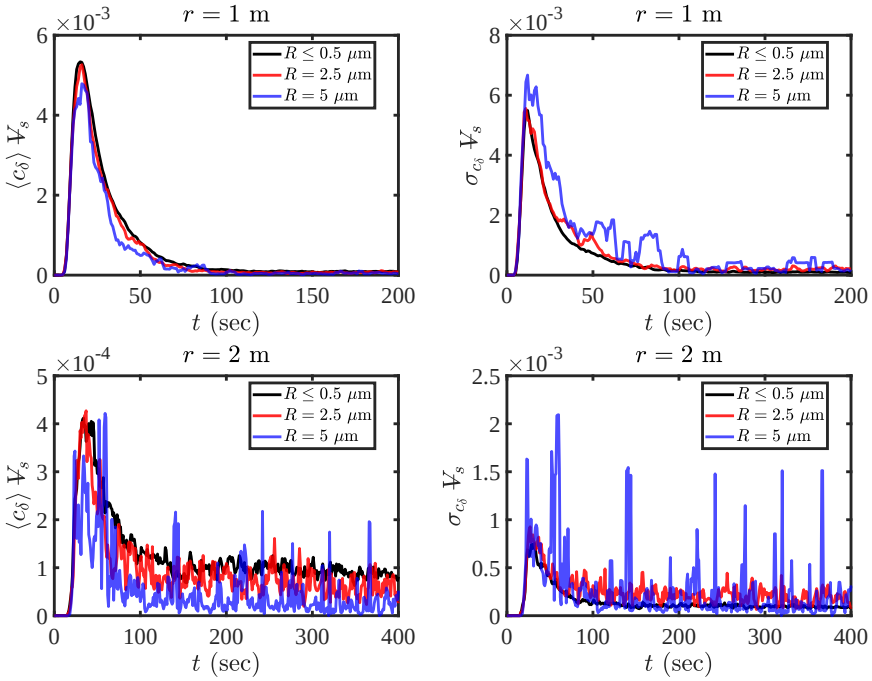


Figure 12: Temporal evolution of (A,C) $\langle c_\delta\rangle$ and (B,D) σ_{c_δ} for (A,B) $r = 1$ m and (C,D) $r = 2$ m for particles of radii, $R \leq 0.5\,\mu$m, $R = 2.5\,\mu$m and $R = 5\,\mu$m.

The guidelines presented so far in this study are for tracer particles of negligible inertia and settling. However, real scenarios may involve turbulent dispersal of particles spanning a wide range of sizes. This subsection will consider larger inertial particles and explore how the criteria developed for the

Journal Pre-proof

tracer particles can be extended to obtain minimum conditions for $\langle \mathcal{N}_{so} \rangle$ and $\mathcal{R}$ that apply for the larger particles.

The process of emission at the source and the process of observation of the pollutant particles at the sink remain the same for both tracer and inertial particles. Only the turbulent transport process by the indoor flow and gravitational settling are different. Therefore, Eq. (9) can be used as the guideline for $\mathcal{R}\langle \mathcal{N}_{so} \rangle$ even for inertial particles. As the particle size increases, particle deposition by settling increases and a lower fraction of particles released from the source reaches the sink. Changes to the minimum required value of $\mathcal{R}\langle \mathcal{N}_{so} \rangle$ for larger particle sizes arise primarily due to this reduction in the fraction of particles reaching the sink. If the reduction in c_δ is considerable, the choice of $\tilde{p} = 0.002$ should be correspondingly lowered, which will result in a higher requirement for $\mathcal{R}\langle \mathcal{N}_{so} \rangle$.

Figure 12 plots $\langle c_\delta \rangle V_s$ for the source-sink configuration shown in Figure 1 at $r = 1$ and 2 m for three particle sizes. A particle radius, $R \leq 0.5\,\mu$m, corresponds to the tracer limit. For larger particle sizes, specifically $R = 2.5$ and $5\,\mu$m, the statistics were evaluated over $\mathcal{R} = 160$ turbulent realizations with $\langle \mathcal{N}_{so} \rangle = 3748$ and 2213 respectively. The increasingly noisier nature of the curves at $r = 2$ m can be primarily attributed to the lower values of $\langle \mathcal{N}_{so} \rangle$. As anticipated, the magnitude of $\langle c_\delta \rangle V_s$ decreases as particle size increases.

The minimum requirement for the number of turbulent realizations, $\mathcal{R}$, can be obtained from Eqs. (11) and (12). Figure 12 also plots $\sigma_{c_\delta} V_s$ as a function of time for the same 3 particle sizes at $r = 1$ and 2 m. Unlike $\langle c_\delta \rangle V_s$, which follows a clear trend of reduction in magnitude with an increase in particle size, we do not observe a similar obvious trend for $\sigma_{c_\delta} V_s$. Fitting the DFRs to the modified Birnbaum-Saunders function for $R = 2.5$ and $5\,\mu$m at $r = 1$ m yields $\max\left[\dfrac{\sigma_A}{\langle A \rangle}, \dfrac{\sigma_\mu}{\langle \mu \rangle}, \dfrac{\sigma_\gamma}{\langle \gamma \rangle}\right] = 0.396$ and 0.48, respectively. To capture the statistics of $\langle c_\delta \rangle$ at $\alpha = \beta = 0.05$, we obtain minimum requirements of 242 and 356 turbulent realizations respectively. This is higher than the minimum requirement of 216 turbulent realizations we obtained for the tracer limit. To summarize, the criteria derived in the earlier sections apply well for inertial particles as well. However, the increased effect of settling will increase the minimum requirement of $\langle N_{so} \rangle$ and $\mathcal{R}$ to achieve the same level of accuracy.

5. Conclusions

In this article, we define the fine-grain delta function response (DFR) as the pollutant concentration at a sink following a delta function emission at the source in one particular realization. The ensemble-averaged DFR and the two-time correlation of the DFR fluctuation are then defined as fundamental

Journal Pre-proof

quantities. These fundamental quantities are used to express the mean and standard deviation of pollutant concentration in any problem of deterministic or stochastic pollutant emission process. We numerically evaluate the DFR using Lagrangian trajectories of pollutant particles, modeled in Euler-Lagrange simulations, as they travel from an emission source to a sink location. We address the question of how to design the Euler-Lagrange simulations and determine the numerical parameters in order to evaluate the ensemble average and two-time correlation of the DFR to any desired level of accuracy.

The Euler-Euler (EE) approach [38] has also been used to study pollutant dispersal. In the current context, the EE approach can be applied to extract the DFR, offering an alternative to the EL approach. However, with the Eulerian approach, when it comes to evaluating the DFR of an ensemble of source-sink combinations, each source release must be considered as an independent pollutant concentration scalar field [35, 52, 53]. Solving the time evolution of an ensemble of advection-diffusion equations is computationally very expensive. The EL approach offers a convenient workaround with its ability to keep track of all the pollutant particles that originated at the source and reached the sink.

This approach of tracking pollutant particles from a source to a sink clearly depends on the particular locations of the chosen source-sink combination. This raises the formidable challenge of how to investigate numerous different source-sink combinations to fully analyze the risk of pollutant dispersal in a computationally efficient manner - note that conducting separate simulations for each source-sink combination is computationally infeasible. The one-way coupled nature of the problem, along with the ability to precisely track the source of each pollutant in the Lagrangian approach, allows us to evaluate the dispersal between many different source-sink combinations with a single simulation, in a computationally efficient manner. We call this the *statistical overloading approach* since the number of pollutant particles within the computational domain will be far in excess of what is observed in any real experiment. By computing the DFR of many different source-sink combinations, we are able to obtain spatially averaged results (homogenization) as well as spatial variability in the pollutant concentration.

We apply the statistical overloading technique in the context of pollutant dispersal in a ventilated indoor space, using Euler-Lagrange large eddy simulations. In such pollutant dispersal Euler-Lagrange simulations, one must make well-informed choices on key computational parameters: (i) the volume of the discrete source/sink, V_s, (ii) the number of independent realizations $\mathcal{R}$ needed to evaluate aleatoric uncertainty associated with turbulent fluctuations, (iii) the minimum number of particles that must be emitted at the source on average in each realization, $\langle \mathcal{N}_{so} \rangle$, (iv) the sampling window for approximating

Journal Pre-proof

the discrete delta function release, Δt, and (v) the sampling frequency at the sink, δt. By analyzing the statistical role of these computational parameters, we seek to provide guidelines that can be applied to any implementation of the statistical overloading framework for the investigation of turbulent dispersal.

The primary role of the source/sink volume and sampling window is to discretize the delta function emission for a single turbulent realization. Smaller values of V_s and Δt will require far too many particles to be tracked; on the other hand, larger values make DFR too coarse both spatially and temporally. In the present problem of indoor pollution, we have chosen V_s to be such that there are 5000 source/sink sub-volumes within the room and Δt to be of the order of a few minutes. Their values must be judiciously chosen depending on the problem setting.

From statistical arguments, to evaluate the mean DFR, $\langle c_\delta \rangle$ and the two-time correlation $\langle c'_\delta(t_1) c'_\delta(t_2) \rangle$, to an of accuracy of $(1-\beta)\%$ with $(1-\alpha)\%$ confidence, we obtain the following computational requirements

1. $\delta t < V_s^{1/3}/u_{\max}$, where $u_{\max}$ is the maximum local velocity at the source/sink.
2. $\mathcal{R}\langle \mathcal{N}_{so} \rangle > \max_t \left\{ \dfrac{Z^2_{\alpha/2}}{\beta^2} \dfrac{1}{\langle c_\delta \rangle V_s} \dfrac{\sigma^2_{c_\delta}}{\langle c_\delta \rangle^2} \right\}$
3. $\sqrt{\mathcal{R}} > \max \left[\dfrac{\sigma_A}{\langle A \rangle}, \dfrac{\sigma_\mu}{\langle \mu \rangle}, \dfrac{\sigma_\gamma}{\langle \gamma \rangle} \right] \dfrac{t_{\alpha/2,\mathcal{R}-1}}{\beta}$
4. $\dfrac{\chi_1^2}{\mathcal{R}-1} \leq \left\{ \dfrac{\sigma_A^2}{\sigma_{A,tr}^2}, \dfrac{\sigma_\mu^2}{\sigma_{\mu,tr}^2}, \dfrac{\sigma_\gamma^2}{\sigma_{\gamma,tr}^2} \right\} \leq \dfrac{\chi_2^2}{\mathcal{R}-1}$, where $\chi_1^2 = \chi_{\alpha/2,\mathcal{R}-1}$ and $\chi_2^2 = \chi_{(1-\alpha/2,\mathcal{R}-1)}$

Guidelines (2) and (3) help determine the minimum values of $\mathcal{R}$ and $\langle \mathcal{N}_{so} \rangle$ required to accurately capture the mean DFR $\langle c_\delta \rangle$. Guidelines (2) and (3) require a choice to be made for the ratio of the standard deviation to the mean of the quantities of interest to determine the minimum values for $\mathcal{R}$ and $\mathcal{N}_{so}$. From the analyses, a value of 0.5 for this ratio represents a reasonable choice that covers over 95 % of all the dispersal scenarios. Guideline (4) determines the minimum values of $\mathcal{R}$ required to accurately capture the two-time correlation. As can be expected, guideline (4) for the higher-order statistics is more stringent than (3) for the mean.

For homogenization based on separation distance, the following guidelines specify the minimum number of source-sink combinations, $\mathcal{R}_s$, required for accurate estimation of the homogenized mean and variance.

Journal Pre-proof

5. $\sqrt{\mathcal{R}_s} > \max\left[\dfrac{\sigma_A}{\langle A\rangle}, \dfrac{\sigma_\mu}{\langle\mu\rangle}, \dfrac{\sigma_\gamma}{\langle\gamma\rangle}\right] \dfrac{Z_{\alpha/2}}{\beta}$

6. $\sqrt{\mathcal{R}_s} > \dfrac{\sigma_{\langle\sigma_{n,1}\rangle}}{\langle\!\langle\sigma_{n,1}\rangle\!\rangle} \dfrac{Z_{\alpha/2}}{\beta}$

7. $\dfrac{\chi_1^2}{\mathcal{R}_s - 1} \leq \dfrac{\langle\!\langle\sigma_{n,2}\rangle\!\rangle^2}{\langle\!\langle\sigma_{n,2}\rangle\!\rangle_{tr}^2} \leq \dfrac{\chi_2^2}{\mathcal{R}_s - 1}$

Guidelines (5), (6) and (7) can be used to determine $\mathcal{R}_s$, $\langle\!\langle c_\delta\rangle\!\rangle$, $\langle\!\langle c'_\delta(t_1)\, c'_\delta(t_2)\rangle\!\rangle$, and $\langle\langle c_\delta\rangle'(t_1)\, \langle c_\delta\rangle'(t_2)\rangle$ to the desired level of accuracy and confidence. Using Guideline (7), we obtain a requirement of $\mathcal{R}_s > 770$. However, due to the non-normal nature of the sample distribution, the applicability of Guideline (7) needed to be verified. Upon using a frequentist resampling approach, we can refine the requirement to $\mathcal{R}_s > 2725$ in order to evaluate $\langle\langle c_\delta\rangle'(t_1)\, \langle c_\delta\rangle'(t_2)\rangle$ at 95% accuracy and confidence.

To summarize the results showcased in this manuscript, the accuracy of the above criteria was evaluated in the present problem of indoor pollutant dispersal. To evaluate all of our quantities of interest at 95% accuracy and 95% confidence, we observe that we need $\mathcal{R}\langle N\rangle_{so} > 192040$, $\mathcal{R} > 770$ and $\mathcal{R}_s > 2725$. When these specific conditions are satisfied, we achieve the desired level of accuracy. Of course, as the level of accuracy and confidence are changed, the requirements change as well.

An important objective of this manuscript is to provide sufficient statistical information that the statistical overloading technique can be easily employed in other turbulent dispersal applications as well. In principle, any flow solver with Lagrangian particle tracking capability should be able to implement the statistical overloading technique. To design a statistically overloaded simulation, first, one would have to choose Δt and V_s based on prior knowledge of the fluid mechanics of the dispersal problem. Following that, a decision on the sampling frequency δt can be made based on Guideline (1). At the scale of a single source-sink combination, a minimum number of turbulent realizations $\mathcal{R}$ needed for statistical convergence should be chosen based on Guideline (3) or (4) depending whether the intended goal is to evaluate just $\langle c_\delta\rangle$ or both $\langle c_\delta\rangle$ and σ_{c_δ}, respectively. Based on the choice of $\mathcal{R}$, Guideline (2) can be used to determine the number of particles that must pass through the source volume, $\mathcal{N}_{so}$, to obtain converged results to desired accuracy and confidence. Finally, to evaluate the double-averaged quantities, Guidelines (5), (6) or (7) can be similarly used to determine the minimum number of source-sink combinations $\mathcal{R}_s$ that must be sampled, depending on the intended goal. In summary, the statistical overloading framework, along with the above criteria, can be used

Journal Pre-proof

to accurately calculate pollutant dispersal in any other turbulent dispersal application.

The statistical overloading approach and methods of parameter selection described in this work were illustrated for computational simulations. The methods could also be applied to experiments, leveraging recent advancements in time-resolved, three-dimensional particle tracking techniques. This would enable experimental characterization of the delta function response for arbitrary release locations and would avoid the limitations of passive scalar injection from wall ports, isokinetic injectors, or other intrusive mechanisms. Modern particle tracking algorithms enable seeding densities appropriate for the methods described here [54]. However, increases in particle track lengths approaching integral timescales of the flow are likely needed to fully adopt the methods to direct measurements of Lagrangian particle trajectories for widely separated source and sink locations. Alternative experimental implementations might instead integrate virtual fluid particle trajectories through the time-resolved velocity fields, which would require quantification of velocity component uncertainty on the derived delta function response. We recommend this as an avenue of future research. In summary, both continued algorithmic advancements and investigation of measurement uncertainties could extend the highly flexible statistical overloading approach to experimental applications.

Appendix A. Stochastic Emission Process

In this appendix, we provide a detailed probabilistic description of the emission process. Let the stochastic emission process be described by the M-time probability density function (PDF) $P_{M,q_{so}}(\zeta_1, t_1, \zeta_2, t_2, \cdots, \zeta_M, t_M, R)$, where $P_{M,q_{so}}\, d\zeta_1, d\zeta_2, \cdots, d\zeta_M$ is the probability of $q_{so}(t, R)$ taking a value within the interval $\zeta_j - (d\zeta_j/2)$ and $\zeta_j + (d\zeta_j/2)$, at time t_j $(j = 1, 2, \cdots, M)$ for pollutants of size R. This is a joint PDF in ζs and therefore satisfies the condition

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} P_{M,q_{so}}(\zeta_1, t_1, \cdots \zeta_M, t_M, R)\, d\zeta_1 \cdots d\zeta_M = 1\,. \tag{A.1}$$

From the above M-time joint PDF, the one-time marginal PDF $P_{1,q_{so}}(\zeta_1, t_1, R)$ can be readily obtained by integrating over all possible emission values at other times. Similarly, the two-time marginal PDF $P_{2,q_{so}}(\zeta_1, t_1, \zeta_2, t_2, R)$ can be readily obtained. The one-time marginal PDF satisfies the condition $\int_{-\infty}^{\infty} P_{1,q_{so}}(\zeta_1, t_1, R)\, d\zeta_1 = 1$. From prior notation of the emission process, we also have the following two definitions

$$\langle q_{so}\rangle(t, R) = \int_{-\infty}^{\infty} \zeta\, P_{1,q_{so}}(\zeta)\, d\zeta \quad \text{and} \quad \sigma^2_{q_{so}}(t, R) = \int_{-\infty}^{\infty} \zeta^2\, P_{1,q_{so}}(\zeta)\, d\zeta - \langle q_{so}\rangle^2\,, \tag{A.2}$$

Journal Pre-proof

where we have suppressed the dependence on t and R on the right hand sides.

Appendix B. Stochastic Transport - Discrete Delta Function Response

We now consider the stochastic transport of pollutant from the source to the sink location by prevailing turbulent flow. For this, we consider a unit emission within a small volume V_s around the source location $\mathbf{x}_{so}$ at time $t = 0$. We then monitor the pollutant concentration within a small volume V_s around the sink location $\mathbf{x}_{si}$ at a later time $t > 0$. In an Eulerian framework, this corresponds to the initial value problem

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{v} c) = 0\,, \quad \text{Initial condition: } c(\mathbf{x}, t = 0) = \frac{1}{V_s}\,, \text{ if } \mathbf{x} - \mathbf{x}_{so} \in V_s \quad \text{(B.1)}$$

where $\mathbf{v}(\mathbf{x}, t)$ is the particle velocity field, which reduces to the turbulent fluid velocity in the limit of tracer-like pollutants. The discrete delta function response (DFR) can then be expressed as the average pollutant concentration around the sink location as

$$c_\delta(t, \mathbf{x}_{so}, \mathbf{x}_{si}) = \frac{1}{V_s} \oint_{V_s} c(\mathbf{x}, t)\, d\mathbf{x}\,. \quad \text{(B.2)}$$

Appendix C. Pollutant Concentration in terms of Stochastic Emission and Transport

We start with a detailed probabilistic description of the turbulent transport processes. Let the M-time turbulent transport process of a delta-function emission be described by the probability density function (PDF) $P_{M,c_\delta}(\xi_1, t_1, \cdots, \xi_M, t_M, \mathbf{x}_{so}, \mathbf{x}_{si}, R)$, where $P_{M,c_\delta}\, d\xi_1 \cdots d\xi_M$ is the joint probability of c_δ taking a value within the interval $\xi_j - (d\xi_j/2)$ and $\xi_j + (d\xi_j/2)$ at time t_j (for $j = 1 \cdots M$) while transporting pollutant of size R, between the source and sink located at $\mathbf{x}_{so}$ and $\mathbf{x}_{si}$, respectively. Since P_{M,c_δ} is a probability in ξs, it satisfies the condition

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} P_{M,c_\delta}(\xi_1, t_1, \cdots, \xi_M, t_M, \mathbf{x}_{so}, \mathbf{x}_{si}, R)\, d\xi_1 \cdots d\xi_M = 1\,. \quad \text{(C.1)}$$

From the M-time joint PDF, the one-time marginal PDF, $P_{1,c_\delta}(\xi_1, t_1)$, and the two-time marginal PDF, $P_{2,c_\delta}(\xi_1, t_1, \xi_2, t_2)$ can be derived by integrating over all possible transport values at the remaining times. From the prior notation of the DFR process, we have the following two definitions

$$\langle c_\delta \rangle(t) = \int_{-\infty}^{\infty} \xi\, P_{1,c_\delta}(\xi)\, d\xi \quad \text{and} \quad \langle c_\delta^2 \rangle = \int_{-\infty}^{\infty} \xi^2\, P_{1,c_\delta}(\xi)\, d\xi\,, \quad \text{(C.2)}$$

Journal Pre-proof

where we have suppressed the dependence on t, $\mathbf{x}_{so}$, $\mathbf{x}_{si}$, and R.

We now proceed to combine the emission and transport processes to obtain appropriate expressions for the mean and standard deviation of pollutant concentration. In the above, when the quantities being averaged are dependent only on the emission or the transport process, the ensemble average reduces to integration over only the emission or the transport probabilities. However, the ensemble average of quantities that depend both on the emission and the transport processes will involve integration over both probabilities. In particular, the ensemble average of pollutant concentration at time t at the sink due to emission that started at $t = 0$ at the source can be expressed as a convolution integral

$$\langle c \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R) = \langle \int_0^t \zeta_1 \xi_{\tilde{1}} \, dt_1 \rangle = \int_0^t \langle \zeta_1 \xi_{\tilde{1}} \rangle \, dt_1 \,, \tag{C.3}$$

where the emission occurs at time t_1 (i.e., $\zeta_1 = \zeta(t_1)$) and is transported to the sink over the time interval $t - t_1$ ($\xi_{\tilde{1}} = \xi(t - t_1)$). Note that the tilde over 1 (i.e., $\tilde{1}$) is used to denote $(t - t_1)$. The kernel of the integration can be expressed as

$$\langle \zeta_1 \xi_{\tilde{1}} \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \zeta_1 \, P_{q_{so}}(\zeta_1, t_1) \, \xi_{\tilde{1}} \, P_{c_\delta}(\xi_{\tilde{1}}, t - t_1) \, d\xi_{\tilde{1}} \, d\zeta_1 \,. \tag{C.4}$$

We then make the substitution $\zeta_1(t) = \langle q_{so} \rangle(t, R) + \zeta_1'(t)$ and $\xi_{\tilde{1}}(t) = \langle c_\delta \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R) + \xi_{\tilde{1}}'(t)$ to obtain the relation

$$\langle c \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R) = \int_0^t \langle q_{so} \rangle(t_1) \, \langle c_\delta \rangle(t - t_1) \, dt_1 \,. \tag{C.5}$$

Where we have used $\langle \zeta_1' \rangle = 0$ and $\langle \xi_{\tilde{1}}' \rangle = 0$. In addition, we have made the reasonable assumption that the transport by turbulence is independent of the emission process and as a result, the correlation between the two processes is zero, i.e., $\langle \zeta_1' \, \xi_{\tilde{1}}' \rangle = 0$. In other words, we have assumed the emitted pollutants to be one-way coupled with the flow. So, the pollutants are advected by the turbulent flow, but they do not alter the flow by their presence.

We now proceed to express the ensemble-averaged mean square concentration in the following way

$$\langle c^2 \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R) = \langle \left(\int_0^t \zeta_1 \, \xi_{\tilde{1}} \, dt_1 \right) \left(\int_0^t \zeta_2 \, \xi_{\tilde{2}} \, dt_2 \right) \rangle = \int_0^t \int_0^t \langle \zeta_1 \, \xi_{\tilde{1}} \, \zeta_2 \, \xi_{\tilde{2}} \rangle \, dt_1 \, dt_2 \,, \tag{C.6}$$

where concentration square is obtained as the product of (i) emission ζ_1 at time t_1 at the source followed by the fraction $\xi_{\tilde{1}}$ advected to the sink over the

time interval $t - t_1$ integrated over all possible $0 \leq t_1 \leq t$ and (ii) emission ζ_2 at time t_2 followed by the fraction $\xi_{\tilde{2}}$ advected to the sink over the time interval $t - t_2$ integrated over $0 \leq t_2 \leq t$. The ensemble average of the two time emission and transport can be written as

$$\langle \zeta_1 \, \xi_{\tilde{1}} \, \zeta_2 \, \xi_{\tilde{2}} \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \zeta_1 \zeta_2 \, P_{2,q_{so}}(\zeta_1, \zeta_2, t_1, t_2) \\ \xi_{\tilde{1}} \xi_{\tilde{2}} \, P_{2,c_\delta}(\xi_{\tilde{1}} \xi_{\tilde{2}}, t - t_1, t - t_2) \, d\zeta_1 \, d\zeta_2 \, d\xi_{\tilde{1}} \, d\xi_{\tilde{2}} \, , \tag{C.7}$$

where $P_{2,q_{so}}(\zeta_1, \zeta_2, t_1, t_2)$ is the probability of emission being ζ_1 and ζ_2 at times t_1 and t_2, respectively and $P_{2,c_\delta}(\xi_{\tilde{1}} \xi_{\tilde{2}}, t - t_1, t - t_2)$ is the probability of delta emission's transport over time intervals $t - t_1$ and $t - t_2$ being $\xi_{\tilde{1}}$ and $\xi_{\tilde{2}}$, respectively.

On the right hand side of (C.6), in the ensemble average $\langle \zeta_1 \, \xi_{\tilde{1}} \, \zeta_2 \, \xi_{\tilde{2}} \rangle$ we now substitute

$$\begin{aligned} \zeta_1 &= \langle q_{so} \rangle(t_1, R) + \zeta_1' \, , \quad \xi_{\tilde{1}} = \langle c_\delta \rangle(t - t_1, \mathbf{x}_{so}, \mathbf{x}_{si}, R) + \xi_{\tilde{1}}' \, , \\ \zeta_2 &= \langle q_{so} \rangle(t_2, R) + \zeta_2' \, , \quad \xi_{\tilde{2}} = \langle c_\delta \rangle(t - t_2, \mathbf{x}_{so}, \mathbf{x}_{si}, R) + \xi_{\tilde{2}}' \, , \end{aligned} \tag{C.8}$$

which results in 16 double integrals in time. Since $\langle \zeta_1' \rangle = 0$, $\langle \zeta_2' \rangle = 0$, $\langle \xi_{\tilde{1}}' \rangle = 0$, and $\langle \xi_{\tilde{2}}' \rangle = 0$, 12 of these 16 terms are zero and we obtain the simplified expression

$$\begin{aligned} \langle c^2 \rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R) = &\left(\int_0^t \langle q_{so} \rangle(t_1) \, \langle c_\delta \rangle(t - t_1) \, dt_1 \right) \left(\int_0^t \langle q_{so} \rangle(t_2) \, \langle c_\delta \rangle(t - t_2) \, dt_2 \right) + \\ &\int_0^t \int_0^t \langle q_{so} \rangle(t_1) \, \langle q_{so} \rangle(t_2) \, \underbrace{\langle \xi_{\tilde{1}}' \xi_{\tilde{2}}' \rangle(t - t_1, t - t_2)}_{\substack{\text{2-time transport} \\ \text{fluctuation correlation}}} \, dt_1 \, dt_2 + \\ &\int_0^t \int_0^t \langle c_\delta \rangle(t - t_1) \, \langle c_\delta \rangle(t - t_2) \quad \underbrace{\langle \zeta_1' \zeta_2' \rangle(t_1, t_2)}_{\substack{\text{2-time emission} \\ \text{fluctuation correlation}}} \quad dt_1 \, dt_2 + \\ &\int_0^t \int_0^t \langle \zeta_1' \, \xi_{\tilde{1}}' \, \zeta_2' \, \xi_{\tilde{2}}' \rangle(t_1, t_2) \, dt_1 \, dt_2 \, . \end{aligned} \tag{C.9}$$

Using (C.6), the first term on the right hand side can be expressed as $\langle c \rangle^2(t, \mathbf{x}_{so}, \mathbf{x}_{si}, R)$, which when subtracted from the left hand side yields the variance of pollutant concentration. While the two-time correlations of the emission and transport processes are expected to be non-zero, we expect their joint correlation $\langle \zeta_1' \, \xi_{\tilde{1}}' \, \zeta_2' \, \xi_{\tilde{2}}' \rangle = 0$. This is because the emitted pollutants do

Journal Pre-proof

Journal Pre-proof

not influence the turbulent flow and the emission and transport processes are uncorrelated. Thus, the last term on the right hand side can be neglected. The resulting simplified equation is discussed in the main text.

Acknowledgments

We gratefully acknowledge support from NSF, USA (EAGER Grant no. 2134083), LG Electronics, Korea (Grant no. C2021017165), DOE/NNSA Minority Serving Institutions Partnership Program under Award No. DE-NA0004003, and the University of Florida Informatics Institute, USA.

References

[1] S. Asadi, N. Bouvier, A. S. Wexler, W. D. Ristenpart, The coronavirus pandemic and aerosols: Does covid-19 transmit via expiratory particles?, Aerosol Science and Technology 54 (2020) 635–638.

[2] M. Nicas, Estimating exposure intensity in an imperfectly mixed room, American Industrial Hygiene Association Journal 57 (1996) 542–550.

[3] M. Nicas, An analytical framework for relating dose, risk, and incidence: an application to occupational tuberculosis infection, Risk Analysis 16 (1996) 527–538.

[4] M. Nicas, C. Keil, C. Simmons, T. Anthony, Turbulent eddy diffusion models, Mathematical models for estimating occupational exposure to chemicals (2009) 53–65.

[5] J. Wagner, T. L. Sparks, S. Miller, W. Chen, J. M. Macher, J. M. Waldman, Modeling the impacts of physical distancing and other exposure determinants on aerosol transmission, Journal of Occupational and Environmental Hygiene 18 (2021) 495–509.

[6] C. E. Feigley, J. S. Bennett, E. Lee, J. Khan, Improving the use of mixing factors for dilution ventilation design, Applied occupational and environmental hygiene 17 (2002) 333–343.

[7] A. C. Megri, F. Haghighat, Zonal modeling for simulating indoor environment of buildings: Review, recent developments, and applications, Hvac&R Research 13 (2007) 887–905.

[8] M. Nicas, Modeling turbulent diffusion and advection of indoor air contaminants by markov chains, AIHAJ-American Industrial Hygiene Association 62 (2001) 149–158.

Journal Pre-proof

[9] M. Nicas, The near field/far field model with constant application of chemical mass and exponentially decreasing emission of the mass applied, Journal of Occupational and Environmental Hygiene 13 (2016) 519–528. doi:10.1080/15459624.2016.1148268.

[10] A. Venkatram, J. Weil, Modeling turbulent transport of aerosols inside rooms using eddy diffusivity, Indoor Air 31 (2021) 1886–1895.

[11] B. J. Schimmoller, N. S. Trovão, M. Isbell, C. Goel, B. F. Heck, T. C. Archer, K. D. Cardinal, N. B. Naik, S. Dutta, A. Rohr Daniel, et al., Covid-19 exposure assessment tool (ceat): Exposure quantification based on ventilation, infection prevalence, group characteristics, and behavior, Science Advances 8 (2022) eabq0593.

[12] J. Schwartz, D. W. Dockery, Increased mortality in philadelphia associated with daily air pollution concentrations, Am Rev Respir Dis 145 (1992) 600–604.

[13] B. Sawford, Turbulent relative dispersion, Annual review of fluid mechanics 33 (2001) 289–317.

[14] M. Cassiani, A. Stohl, S. Eckhardt, The dispersion characteristics of air pollution from the world's megacities, Atmospheric Chemistry and Physics 13 (2013) 9975–9996.

[15] P. F. i Soler, Fundamental Numerical Studies of Thermal Stratification Effects in Idealized Urban Areas, Ph.D. thesis, TU Delft, 2023.

[16] R. Naidu, B. Biswas, I. R. Willett, J. Cribb, B. K. Singh, C. P. Nathanail, F. Coulon, K. T. Semple, K. C. Jones, A. Barclay, et al., Chemical pollution: A growing peril and potential catastrophic risk to humanity, Environment International 156 (2021) 106616.

[17] R. E. Britter, S. R. Hanna, Flow and dispersion in urban areas, Annual review of fluid mechanics 35 (2003) 469–496.

[18] R. Langer, Drug delivery and targeting., Nature 392 (1998) 5–10.

[19] A. Folch, A review of tephra transport and dispersal models: Evolution, current status, and future perspectives, Journal of Volcanology and Geothermal Research 235 (2012) 96–115.

[20] K.-C. Cheng, V. Acevedo-Bolton, R.-T. Jiang, N. E. Klepeis, W. R. Ott, O. B. Fringer, L. M. Hildemann, Modeling exposure close to air pollution

Journal Pre-proof

sources in naturally ventilated residences: Association of turbulent diffusion coefficient with air change rate, Environmental science & technology 45 (2011) 4016–4022.

[21] K. Krishnaprasad, N. Zgheib, K. Choudhary, M. Ha, C. Choi, K. Bang, S. Jang, S. Balachandar, et al., Existence of a nonzero worst-case ach for short-term exposure in ventilated indoor spaces, Indoor Air 2024 (2024).

[22] J. S. Salinas, K. Krishnaprasad, N. Zgheib, S. Balachandar, Improved guidelines of indoor airborne transmission taking into account departure from the well-mixed assumption, Physical Review Fluids 7 (2022) 064309.

[23] K. Krishnaprasad, J. Salinas, N. Zgheib, S. Balachandar, Fluid mechanics of air recycling and filtration for indoor airborne transmission, Physics of Fluids 35 (2023).

[24] K. J. Allwine, J. E. Flaherty, Joint Urban 2003: Study overview and instrument locations, Technical Report, Pacific Northwest National Lab.(PNNL), Richland, WA (United States), 2006.

[25] K.-C. Cheng, C.-H. Tseng, L. M. Hildemann, Using indoor positioning and mobile sensing for spatial exposure and environmental characterizations: pilot demonstration of pm2. 5 mapping, Environmental Science & Technology Letters 6 (2019) 153–158.

[26] W. H. Organization, et al., Infection prevention and control during health care when novel coronavirus (nCoV) infection is suspected: interim guidance, 25 January 2020, Technical Report, World Health Organization, 2020.

[27] W. H. Organization, et al., Roadmap to improve and ensure good indoor ventilation in the context of covid-19 (2021).

[28] K. Choudhary, K. Krishnaprasad, N. Zgheib, M. Ha, S. Balachandar, Effect of room size, shape, ac placement, and air leakage on indoor airborne viral transmission, Building and Environment 244 (2023) 110834.

[29] E. Riley, G. Murphy, R. Riley, Airborne spread of measles in a suburban elementary school, American Journal of Epidemiology 107 (1978) 421–432.

[30] M. Z. Bazant, J. W. Bush, A guideline to limit indoor airborne transmission of covid-19, Proceedings of the National Academy of Sciences 118 (2021).

Journal Pre-proof

[31] S. Tan, Z. Zhang, K. Maki, K. J. Fidkowski, J. Capecelatro, Beyond well-mixed: A simple probabilistic model of airborne disease transmission in indoor spaces, Indoor air 32 (2022) e13015.

[32] T. Foat, J. Drodge, J. Nally, S. Parker, A relationship for the diffusion coefficient in eddy diffusion based indoor dispersion modelling, Building and Environment 169 (2020) 106591.

[33] K. Choudhary, K. Krishnaprasad, S. Pandey, N. Zgheib, J. Salinas, M. Ha, S. Balachandar, Effectiveness of rans in predicting indoor airborne viral transmission: A critical evaluation against les, Computers & Fluids 256 (2023) 105845.

[34] R. Patel, K. Krishnaprasad, N. Zgheib, J. S. Salinas, S. Balachandar, Influence of internal heating and natural convection on airborne transmission of viral diseases, Fluid Dynamics Research (2024).

[35] P. Fradera-Soler, P. Johnson, A. Banko, S. Balachandar, Eulerian and lagrangian analysis of dispersion in an idealized urban geometry, Bulletin of the American Physical Society (2024).

[36] A. Shad, H. Ahmed, N. Zgheib, S. Balachandar, S. Sherif, Stokes-dependent droplet collection efficiency on a naca 0012 airfoil: Insights from droplet-informed simulations, Bulletin of the American Physical Society (2024).

[37] S. Balachandar, K. Liu, M. Lakhote, Self-induced velocity correction for improved drag estimation in euler–lagrange point-particle simulations, Journal of Computational Physics 376 (2019) 160–185.

[38] S. Balachandar, Fundamentals of dispersed multiphase flows, Cambridge University Press, 2024.

[39] M. Germano, U. Piomelli, P. Moin, W. H. Cabot, A dynamic subgrid-scale eddy viscosity model, Physics of Fluids A: Fluid Dynamics 3 (1991) 1760–1765.

[40] D. K. Lilly, A proposed modification of the germano subgrid-scale closure method, Physics of Fluids A: Fluid Dynamics 4 (1992) 633–635.

[41] J. Pozorski, J.-P. Minier, On the lagrangian turbulent dispersion models based on the langevin equation, International Journal of Multiphase Flow 24 (1998) 913–945. URL: https://www.sciencedirect.com/science/article/pii/S0301932298000160. doi:https://doi.org/10.1016/S0301-9322(98)00016-0.

Journal Pre-proof

[42] P. Fede, O. Simonin, P. Villedieu, K. Squires, Stochastic modeling of the turbulent subgrid fluid velocity along inertial particle trajectories, in: Proceedings of the Summer Program, Center for Turbulence Research, 2006, pp. 247–258.

[43] K. Liu, P. Huck, A. Aliseda, S. Balachandar, Investigation of turbulent inflow specification in euler–lagrange simulations of mid-field spray, Physics of Fluids 33 (2021) 033313.

[44] A. Dehbi, Turbulent particle dispersion in arbitrary wall-bounded geometries: A coupled cfd-langevin-equation based approach, International Journal of Multiphase Flow 34 (2008) 819–828.

[45] A. A. Mofakham, G. Ahmadi, Particles dispersion and deposition in inhomogeneous turbulent flows using continuous random walk models, Physics of Fluids 31 (2019) 083301.

[46] J. Ferry, S. Balachandar, A fast eulerian method for disperse two-phase flow, International Journal of Multiphase Flow 27 (2001) 1199–1226.

[47] M. O. Deville, P. F. Fischer, E. Mund, et al., High-order methods for incompressible fluid flow, volume 9, Cambridge university press, 2002.

[48] D. Zwick, S. Balachandar, A scalable euler–lagrange approach for multiphase flow simulation on spectral elements, The International Journal of High Performance Computing Applications 34 (2020) 316–339.

[49] J. A. Díaz-García, V. Leiva-Sánchez, A new family of life distributions based on the elliptically contoured distributions, Journal of Statistical Planning and Inference 128 (2005) 445–457.

[50] V. Leiva, The birnbaum-saunders distribution (2015).

[51] L. Villafañe-Roca, M. Esmaily-Moghadam, A. Banko, J. K. Eaton, A robust method for quantification of preferential concentration from finite number of particles, Center for Turbulence Research Annual Research Briefs (2016) 123–135.

[52] Z. Zhang, Q. Chen, Comparison of the eulerian and lagrangian methods for predicting particle transport in enclosed spaces, Atmospheric environment 41 (2007) 5236–5248.

[53] B. Zhao, C. Yang, X. Yang, S. Liu, Particle dispersion and deposition in ventilated rooms: Testing and evaluation of different eulerian and lagrangian models, Building and Environment 43 (2008) 388–397.

Journal Pre-proof

[54] A. Schröder, D. Schanz, 3d lagrangian particle tracking in fluid mechanics, Annual Review of Fluid Mechanics 55 (2023) 511–540.

Journal Pre-proof

Journal Pre-proof

Highlights

- The turbulent dispersal of pollutants can be analyzed using Lagrangian tracking techniques.
- Computational or experimental studies of this problem require careful design choices.
- Guidelines for these design choices are provided using rigorous interval estimation techniques.
- Euler-Lagrange simulations of indoor aerosol dispersal are utilized for validation.
- Guidelines for a novel statistical overloading approach is systematically explored.

Journal Pre-proof

Journal Pre-proof

Declaration of interests

☐ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

K.A. Krishnaprasad reports financial support was provided by National Science Foundation. K.A. Krishnaprasad reports financial support was provided by LG Electronics Inc. Nadim Zgheib reports financial support was provided by National Nuclear Security Administration. Rupal Patel reports financial support was provided by National Science Foundation. Rupal Patel reports financial support was provided by LG Electronics Inc. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Journal Pre-proof